BACSE203
Computer Networks

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Course Objectives

- To foster students understanding of the core concepts of computer networking, including protocols, architectures, and their applications.
- To empower students with the knowledge to design, implement, and analyze the performance of OSI and TCP/IP architectures effectively.
- To identify appropriate application layer protocols for specific applications and their corresponding security mechanisms.

SYLLABUS

Module:1	Networking Foundations	8 hours

Layering and Protocols-Network Topology - Encapsulation -Multiplexing and Demultiplexing - OSI Model, TCP/IP model- Basics of Physical Layer-Internet Architecture-Application Programming Interface (Sockets) - Performance Metrics (Throughput, Bandwidth, Delay and Latency) - High-Speed Networks.

Module:2	Direct Links	9 hours

Connecting to a Network - Encoding – Framing: Byte-Oriented Protocols (PPP), Bit-Oriented Protocols (HDLC); Ethernet (IEEE 802.3); Error Detection: Internet Checksum Algorithm, Cyclic Redundancy Check –Error Correction- Hamming Code; Reliable Transmission: Stop-and-Wait, Sliding Window - Multiple-access Networks: Access Protocol, Wireless Networks: 802.11/Wi-Fi, Bluetooth (802.15.1) - Access Networks, Cellular Network.

Syllabus

Module:3	Internetworking	10 hours

Switching Basics – Datagrams, Virtual Circuit Switching, Virtual LANs (VLANs); Internet (IP) – Internetwork, Service Model, Global Addresses, Datagram Forwarding in IP, Classful addressing, Subnetting and Classless Addressing, Address Resolution Protocol (ARP), Dynamic Host Configuration Protocol (DHCP), Error Reporting-ICMP, Network Address Translation (NAT), Routing - Network as a Graph, Distance-Vector-RIP, Link State-OSPF, Metrics, Implementation-Performance Analysis- Packet Tracer; Global Internet - Routing Areas, Interdomain Routing -BGP; IP Version 6 - Addresses and Routing, Packet Format, Advanced Capabilities; Multicast - Multicast Addresses, Multicast Routing - DVMRP

Syllabus

Module:4	End to End Protocols	8 hours

Simple Demultiplexer (UDP), Reliable Byte Stream (TCP) - End-to-End Issues, Segment Format, Connection Establishment and Termination, Sliding Window, Triggering Transmission, Adaptive Retransmission, Record Boundaries, TCP Extensions, Performance, Alternative Design Choices-SCTP. Remote Procedure Call (RPC)-Fundamentals, Quality of Service -Allocating Resources - Issues in Resource Allocation; Queuing Disciplines - FIFO, Fair Queuing; TCP Congestion Control; Advanced Congestion Control, Quality of Service

Syllabus

Module:5	End to End Data	8 hours

Security Attacks - Trust and Threats, IP Security (IPsec), Firewalls; Applications - Electronic Mail (SMTP, MIME, POP3), World Wide Web (HTTP), FTP, Telnet, Infrastructure Applications-DNS, SNMP; Software Defined Network (SDN)- introduction.

Module:6	Contemporary Issues	2 hours

Industry practices like Netconf and Telemetry

Course Outcomes

On completion of this course, student should be able to:

- Interpret the different building blocks of networks.
- Analyze the error and flow control mechanisms for connecting a network.
- Design and configure efficient networks using switching, IP, routing, NAT multicast, and IPv6, to optimize scalability and performance
- Examine the essential factors for reliable transport layer services, flow control, connection management and analyze the consequences of congestion on their performance.
- Assess the significance of application layer services and their associated protocols for seamless network communication and functionality

Text Book

- Larry L. Peterson, Bruce S. Davie, "Computer Networks : A Systems Approach", The Morgan Kaufmann Series in Networking, 6th Edition, 2021

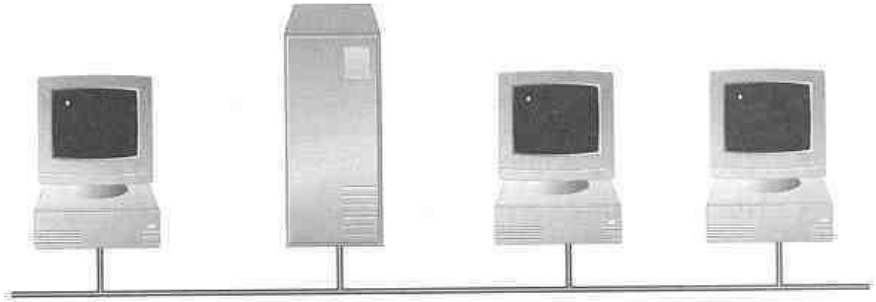
Reference Books

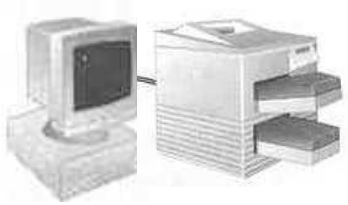
- Behrouz A. Forouzan, "Data communication and Networking", McGraw Hill Education, 6th Edition, 2022
- James F. Kurose and Keith W.Ross, "Computer Networking: A Top-Down Approach", Pearson Education, 9th Edition, 2025
- William Stallings, "Data and Computer Communication", Pearson India Education Services Pvt Ltd, 10th Edition, 2022

What is a Network?

- *A network is a set of devices (often referred to as nodes) connected by communication links.*

- *A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.*





A network consists of 2 or more computers or devices connected together, and they can communicate and share resources (e.g. information)

Computer Network

Definition:

A computer Network is defined as a set of computers interconnected by a communication links for sharing data or resources or exchanging messages.

- A node can be a computer, printer or other device capable of sending and receiving data generated by other nodes on the network.



Communicate

sharing information

Sharing can be local or remote

Local communication

It usually occurs face to face

Remote communication

It takes place over distance

Data Communication

telecommunication

- *The term telecommunication means communication at a distance.*

- *The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data.*

Data communications

- *Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.*

Characteristics of Data Communication

Delivery:

- when a sender transmits data, it must reach only the intended destination and not somewhere

Accuracy:

- The system must deliver the data accurately
- Data have been altered in transmission and left uncorrected are unusable

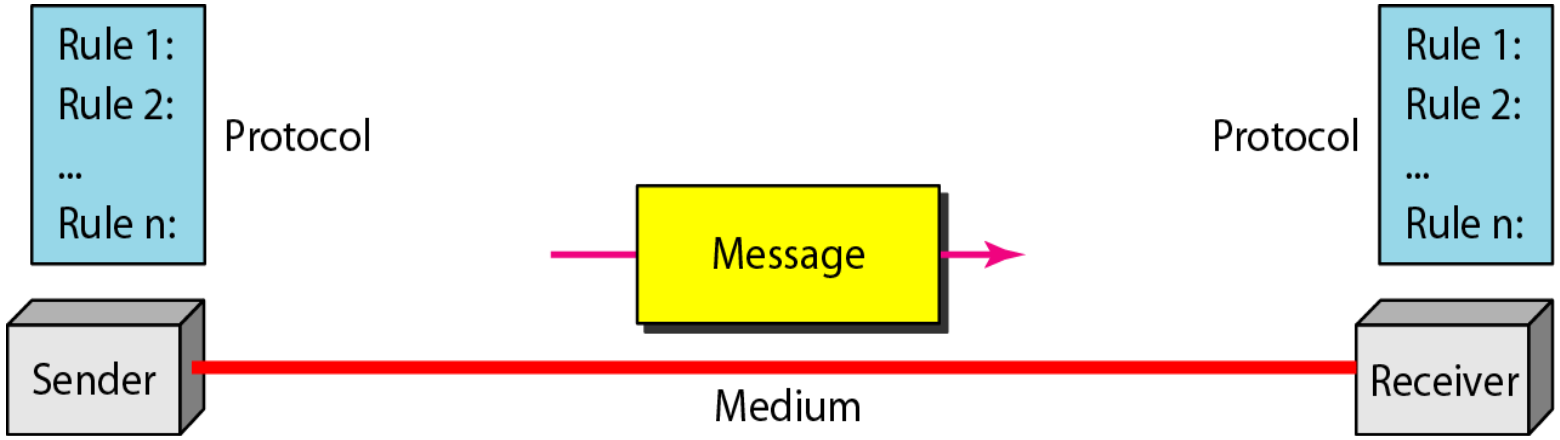
Timeliness:

- The system must deliver data in a timely manner. (i.e) without any delay.

Jitter:

- Jitter refers to the variation in the packet arrival time.
- (i.e) uneven delay in the delivery of audio or video packets

Figure 1.1 *Five components of data communication*



Message: The message is the information (data) to be communicated.
(Ex.) text, numbers, pictures, audio or video

Sender: The sender is the device that sends the data message.
It can be computer, telephone handset, video camera etc.,

Receiver: The receiver is the device that receives the message.
(Ex) Computer, telephone handset, Television etc.,

Transmission Medium: It is physical path by which a message travel from sender to receiver

(Ex) Twisted pair wire, coaxial cable, fiber optic cable or radio waves

Protocol:

- A protocol is a set of rules that govern data communications.
- It represents an agreement between the communicating devices.
- Without a protocol, two devices may be connected but not communicating

(Ex) just as a person speaking French cannot be understood by a person who speaks only Japanese

Network Architecture

What is Network Architecture?

- A computer network must provide general, cost effective and robust connectivity among a large number of computers.
- To deal with this complexity network designers developed the concept of network architecture which introduces a layered protocols or functions that guides the design and implementation of networks
- A **blueprint** for designing and implementing networks.
- Helps manage complexity, ensures scalability, and supports evolving technologies and applications.

Most widely network architectures are

- OSI Architecture
- Internet Architecture

Layering and Protocols

Layering and Protocols

Abstraction

- Abstraction allows developers and users to interact with a system without needing to understand its internal details.
- Example: When you send a message over WhatsApp, you don't need to know how TCP/IP packets are routed - the app abstracts that complexity.
- **Analogy**: Like driving a car with a steering wheel and pedals, without needing to understand the engine mechanics.

Benefits:

- **Simplification** → Easier to design and use networks.
- **Modularity** → Each layer can evolve independently.
- **Flexibility** → Different implementations can coexist as long as they follow the same abstraction rules.

Why Layering?

1) It decomposes the problem of building a network into more manageable components. Each layer solves one part of problem

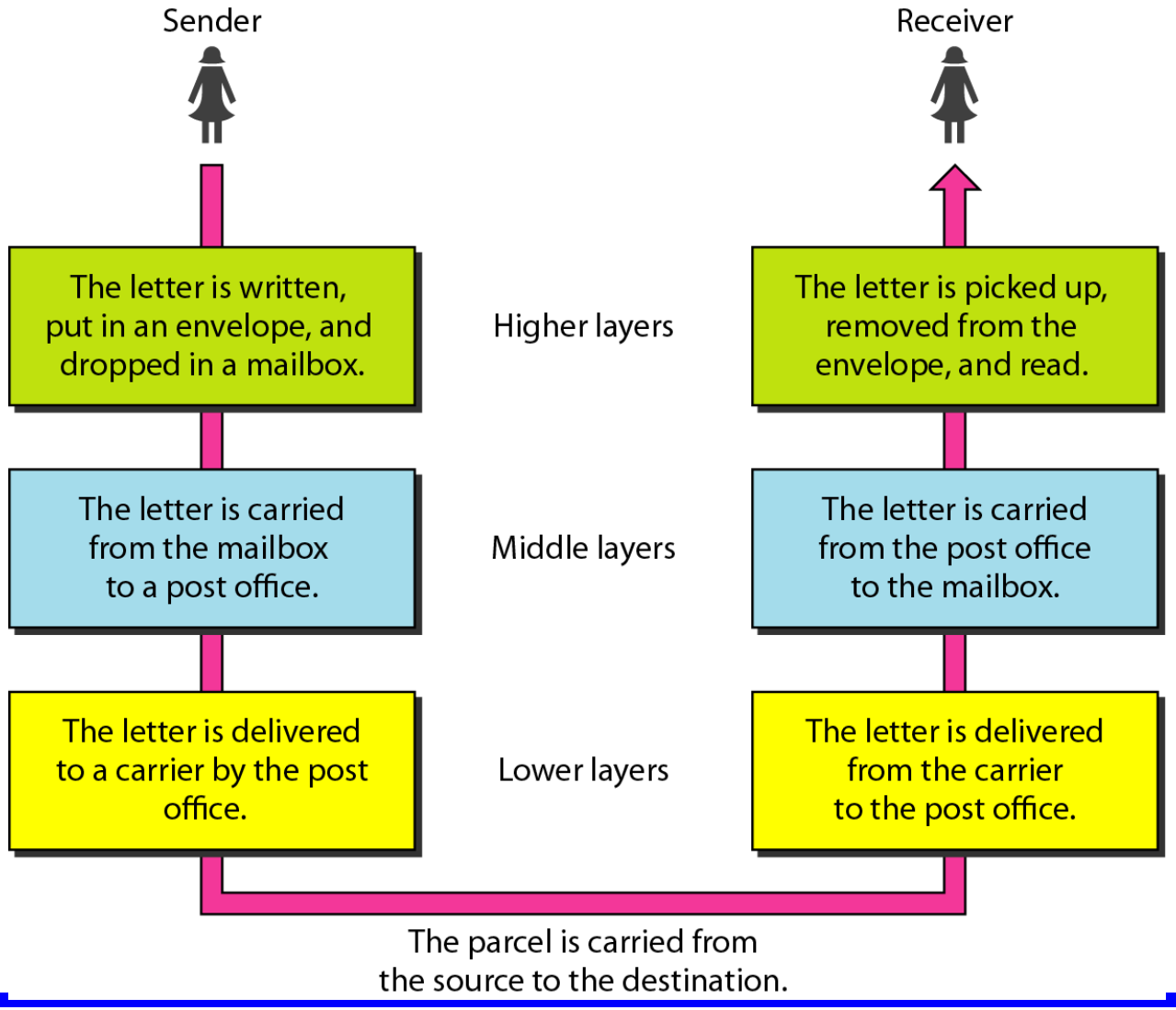
2) It provides a more modular design. That is, if we want to add new service, it requires only modifying the functionality at the respective layers and reusing the functions provided at all the other layers

LAYERED TASKS

- We use the concept of layers in our daily life.

- As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.

Tasks involved in sending a letter



Hierarchy

- According to our analysis, there are three different activities at the sender site and another three activities at the receiver site.

- The task of transporting the letter between the sender and the receiver is done by the carrier.

- Something that is not obvious immediately is that the tasks must be done in the order given in the hierarchy.

- At **the sender site**, the letter must be written and dropped in the mailbox before being picked up by the letter carrier and delivered to the post office.
- At **the receiver site**, the letter must be dropped in the recipient mailbox before being picked up and read by the recipient.

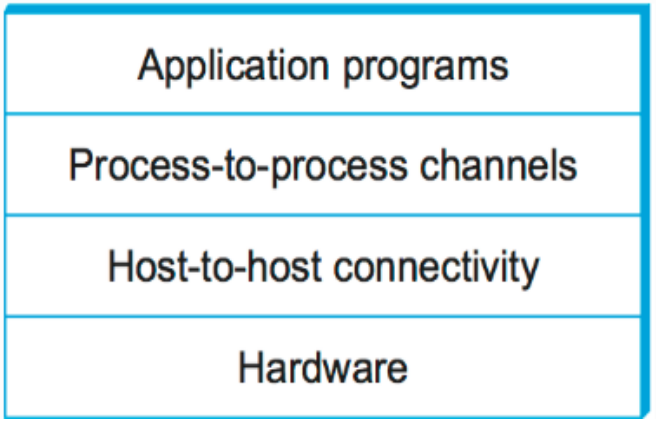
Layering and Protocols

Layering in Networks

- Communication is structured into **layers**, each responsible for specific tasks.

Typical progression:

Hardware layer → raw bit transmission.
Host-to-host layer → ensures reliable connectivity between machines.
Process-to-process layer → enables applications to communicate.

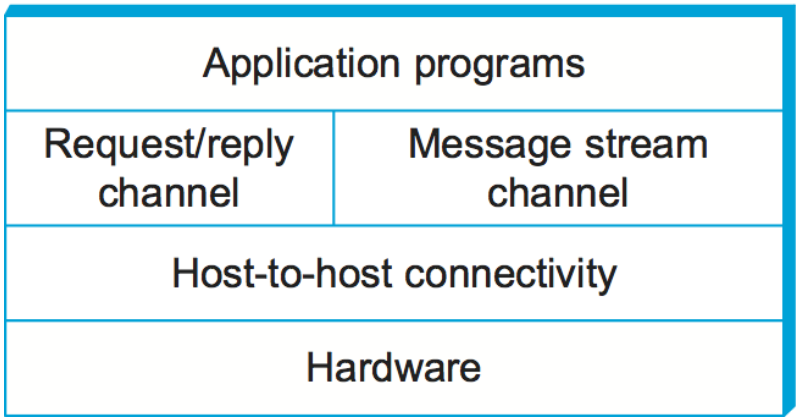


Advantages of layering:

- **Decomposition** → complex problems broken into manageable parts.
- **Modularity** → easier to update or replace one layer without affecting others.
- **Interoperability** → different systems can work together if they follow the same layered rules.

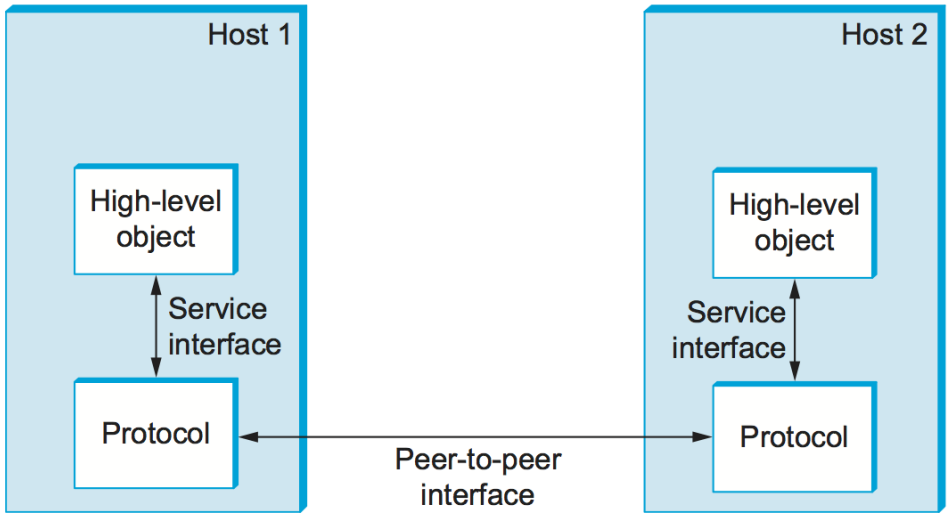
Multiple Abstractions per Layer

- A single layer can support different communication styles:
  - **Request/Reply channel** → like Q&A (used in client-server models).
  - **Message Stream channel** → continuous flow of data (used in video streaming).
- This flexibility allows applications to choose the abstraction that best fits their needs.



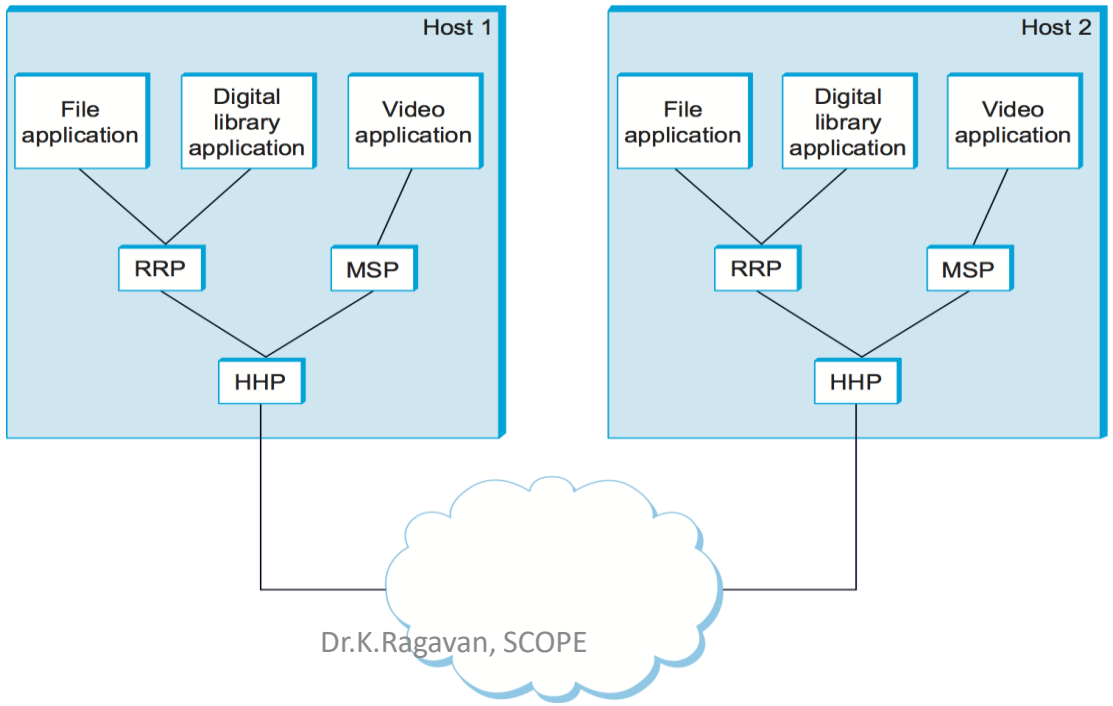
Protocols

- **Definition**: Rules and services that govern communication.
- Two interfaces:
  - **Service interface** → operations available to applications (e.g., HTTP GET).
  - **Peer interface** → rules for communication between machines (e.g., format of HTTP request/response).
- Protocols ensure consistency, predictability, and reliable communication.



Protocol Graphs

- Protocols depend on each other, forming a **protocol graph**.
- Example:
  - RRP (Request/Reply Protocol) and MSP (Message Stream Protocol) both depend on HHP (Host-to-Host Protocol).
- Applications use **protocol stacks** (e.g., RRP + HHP) to achieve end-to-end communication.
- Graphs help visualize dependencies and layering.



Transmission Modes:

Definition: the direction of dataflow between sender to the receiver and vice-versa

Two types:

- Unidirectional
- Bidirectional

Unidirectional

- *Transfer of data from source to destination only*
- *No reverse direction*

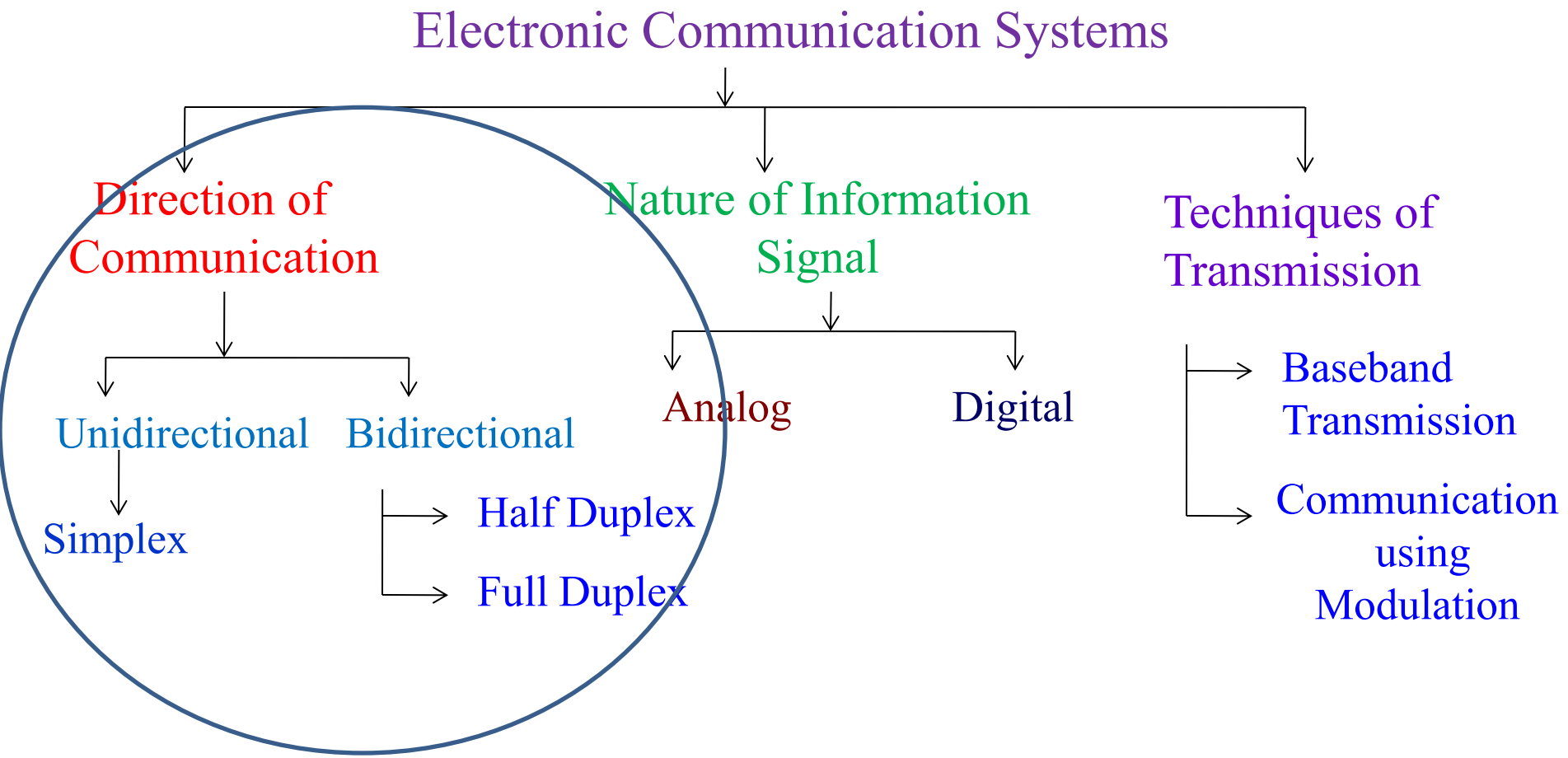
(Ex) Simplex

Bidirectional

- *Two way transmission*

(Ex) Half-Duplex and Full-Duplex

Classification of Electronic Communication Systems



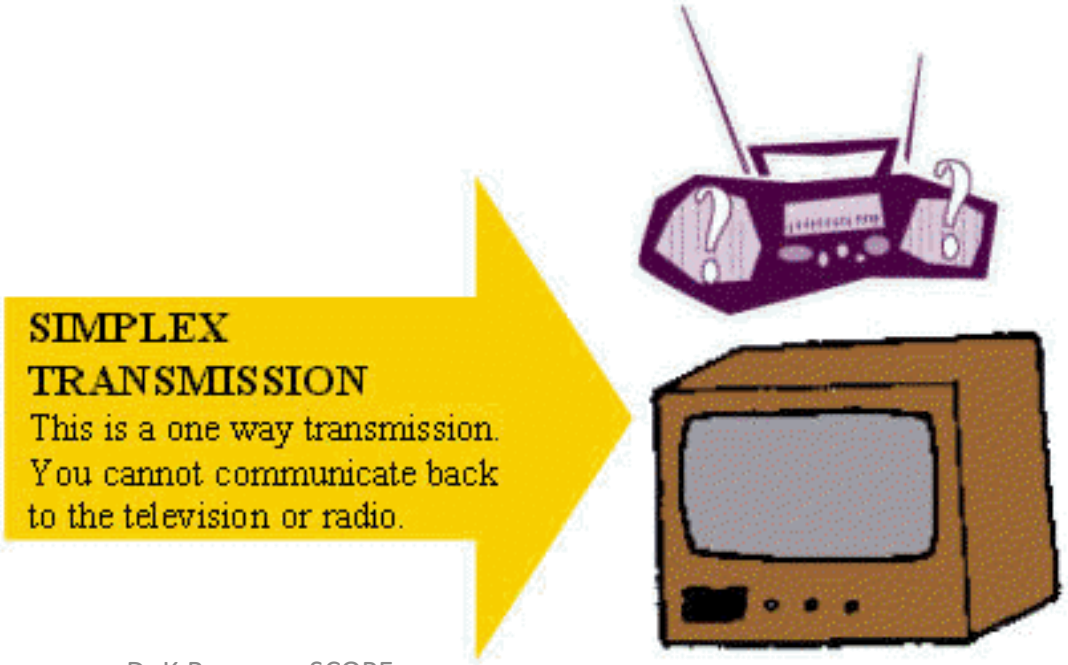
Two basic modes of communication

Broadcasting (or) Unidirectional Communication

Simplex Systems

- Information is communicated in only one direction
- It has Single Powerful Transmitter and Numerous Receivers
- (ex) Radio and Television, Keyboard, monitor





Point- to-Point (or) Bidirectional Communication

Half Duplex

- These system are bidirectional
- They can transmit as well as receive but not simultaneously

  (i.e)
- both directions but only one direction at a time
- It has single transmitter and single receiver

Example

Transreceivers (or) Walky-Talky set



Point- to-Point (or) Bidirectional Communication

Full Duplex

- Truly bidirectional system
- They can transmit as well as receive simultaneously

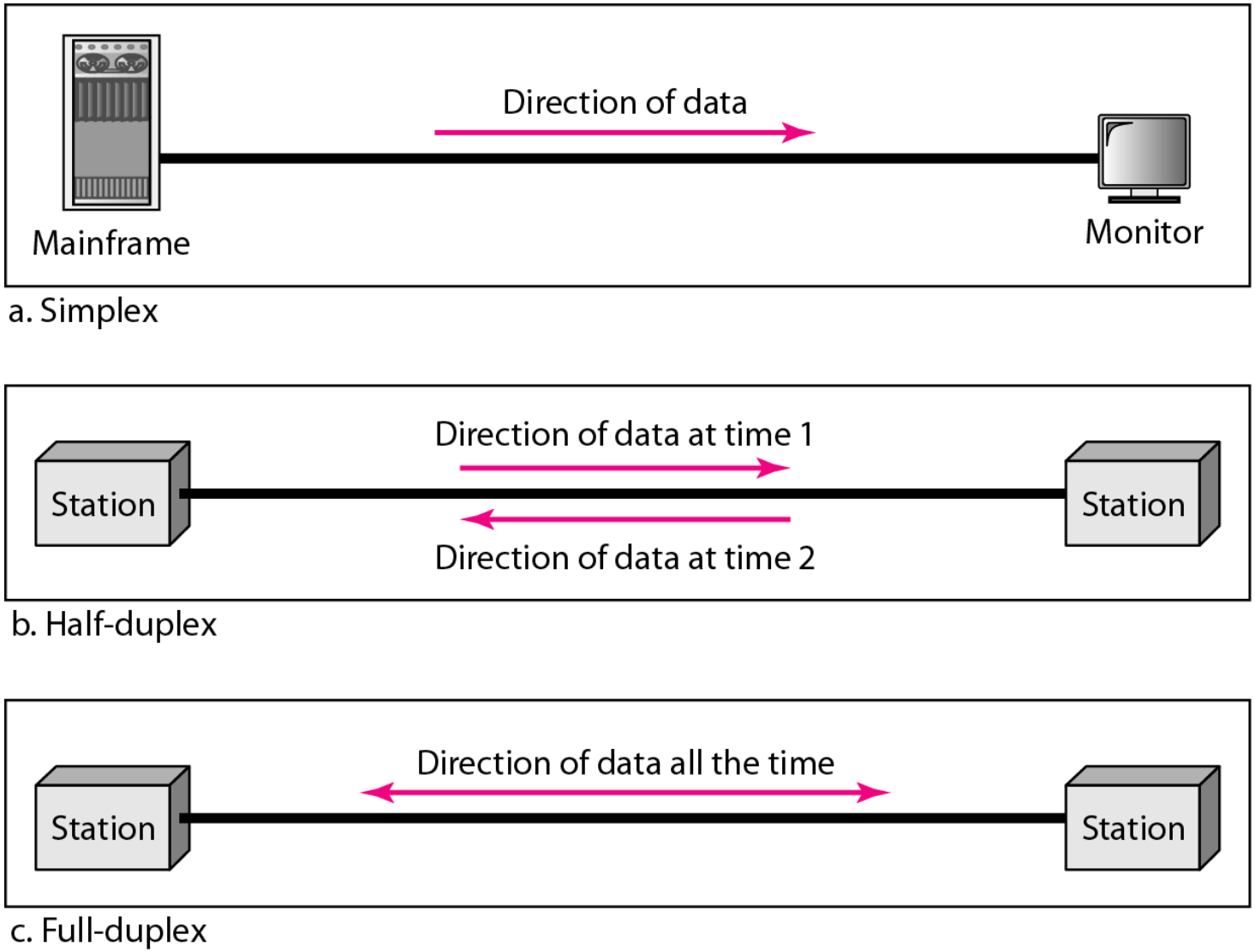
  (i.e)
- Communication can take place in both directions simultaneously
- It has single transmitter and single receiver



Example

- Telephone Systems

Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



Network Criteria

- Network must meet a certain number of criteria

A network must satisfy the following important criteria:

- **Performance:** It depends on the factors such as transit time, response time, and the number of users, type of transmission medium, capabilities of the hardware and the efficiency of the software.

- **Reliability:** Network reliability is measured by the accuracy of delivery, the frequency of failure and the time taken by a link to recover from failure.

- **Security:** It is concerned with the protection of data from unauthorized access.

Advantages of Networking

- Connectivity and Communication
- Data Sharing
- Hardware Sharing
- Internet Access
- Internet Access Sharing
- Data Security and Management
- Performance Enhancement and Balancing
- Entertainment

Applications

- Most people know about the Internet (a computer network) through applications
  - World Wide Web(Reterival of a Web Page)
  - Email
  - Online Social Network
  - Streaming Audio Video(Video on Demand)
  - File Sharing
  - Instant Messaging(Real Time Audio and Video)
  - …

  People:

  Create Application, Manage Networks, Design, build protocols and devices to make the internet.

Encapsulation

Encapsulation

Definition

- Encapsulation is the process of **wrapping application data with protocol-specific control information** (headers or trailers) at each layer of the network stack.
- Each layer adds its own header to instruct its peer how to handle the message.

Message Structure

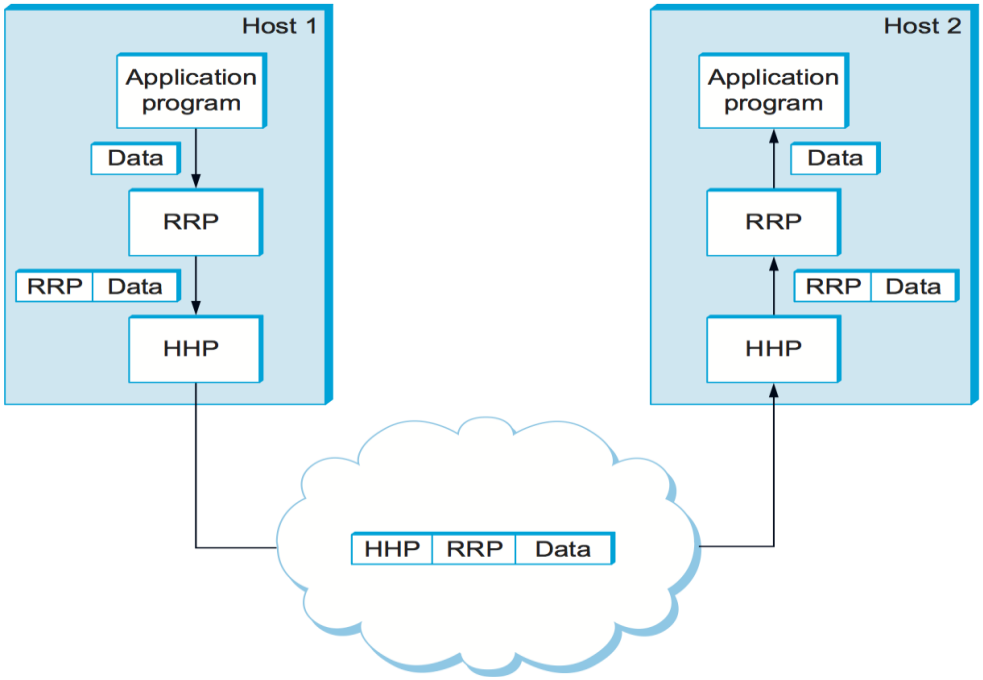
- **Header**: Small data structure (few bytes to dozens) added at the front of the message. Contains control information (e.g., addressing, sequencing, error-checking).
- **Trailer**: Sometimes control information is added at the end of the message.
- **Payload (Body)**: The actual application data (e.g., text, image, video) being transmitted.
- Together, they form a new message at each layer.

Process of Encapsulation

- Application passes raw data (string of bytes) to a protocol (e.g., RRP).
- Protocol attaches its **header** → creates a new message.
- Lower-level protocol (e.g., HHP) encapsulates this message again with its own header.
- At the destination:
  - Each layer **removes its own header** and passes the remaining payload upward.
  - The application finally receives the original data, unchanged.

Example

- Application → sends data to **RRP**.
- RRP attaches its header → passes message to **HHP**.
- HHP attaches its header → sends across the network.
- At destination:
  - HHP reads its header, strips it, and passes payload to RRP.
  - RRP reads its header, strips it, and passes payload to the application.
- The application sees only the original data, not the headers.



Key Points

- **Transparency**: Applications are unaware of headers/trailers added by lower layers.
- **Control Information**: Headers guide peers on how to process messages.
- **Repetition**: Encapsulation happens at every layer of the protocol stack.
- **Decapsulation**: Reverse process at the receiving end.
- **Intermediate Devices**: Routers/switches may inspect certain headers (e.g., HHP/IP headers) for routing decisions.
- **Transformations**: Lower-level protocols may compress or encrypt the entire message (payload + headers).

Benefits of Encapsulation

- Provides **layer independence** → each layer focuses only on its own responsibilities.
- Ensures **modularity** → easy to add/change protocols.
- Supports **security and reliability** → headers can include error detection, encryption, or sequencing info.
- Enables **interoperability** → standardized headers allow different systems to communicate.

Multiplexing and Demultiplexing

Multiplexing and Demultiplexing

Definition

- **Multiplexing**: Combining multiple data flows over a single communication channel or physical link.
- **Demultiplexing**: Separating those flows at the destination and delivering them to the correct application or protocol.

Concept in Networking

- Fundamental idea of **packet switching** → multiple flows share the same physical link.
- Applies across the **protocol stack**, not just at switching nodes.
- Example: RRP (Request/Reply Protocol) can carry messages from multiple applications over one logical channel.

Demultiplexing Key (Demux Key)

- Each protocol attaches an **identifier** in its header to mark which application or higher-level protocol the message belongs to.
- At the destination:
  - Protocol strips its header.
  - Examines the demux key.
  - Passes the message to the correct application or protocol.
- This ensures correct delivery even when multiple applications share the same channel.

Importance

- Enables **resource sharing** → multiple applications can use the same network infrastructure.
- Provides **flexibility** → supports diverse applications simultaneously.
- Ensures **correct delivery** → prevents data from being misrouted.
- Critical for scalability in modern networks.

Network Topology

Physical Structure (or) Scalable Connectivity

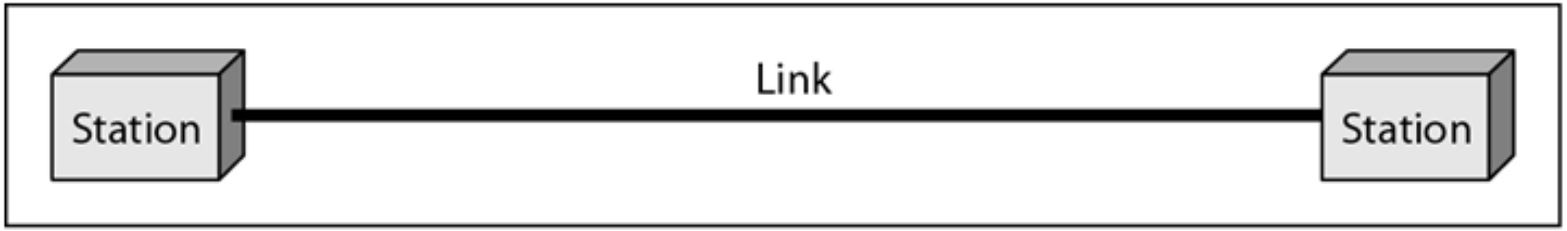
Type of Connections

In a network, two or more devices connected to each other through physical connection is called link, such as coaxial cable or optical fiber.

- **Point to point connection**
- **Multipoint connection**

Point to point connection

- It provides a dedicated direct link between two devices
- Entire capacity of the link is reserved for transmission between those two devices
- Use an actual length of wire or cable



a. Point-to-point

- Other options, such as microwave or satellite is possible

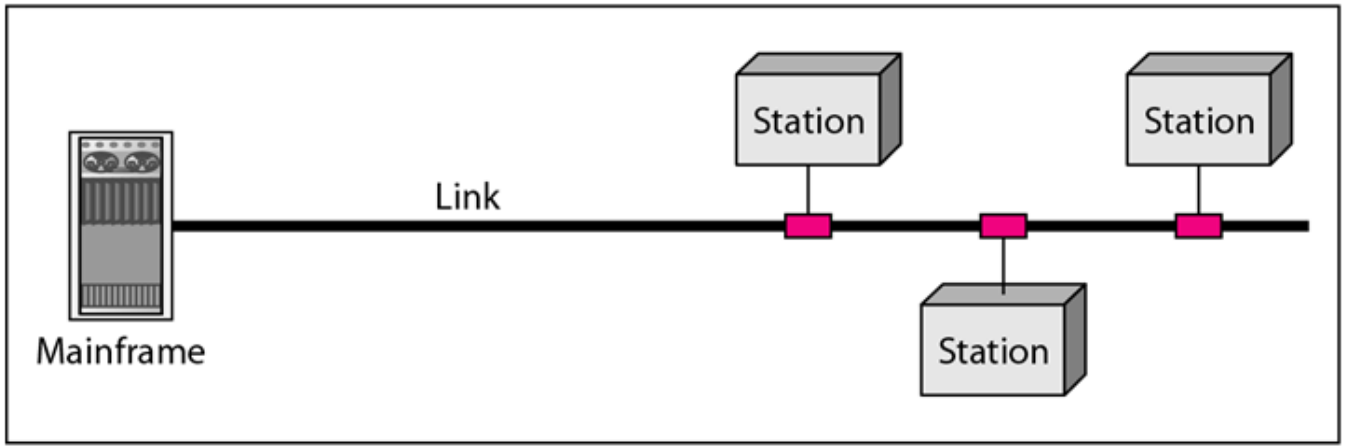
- Ex. Connection between

remote control and TV's control system



Multipoint connection

- More than two devices share a single link
- It supports sharing of channel capacity among stations in the network



b. Multipoint

- Specially shared connection:
  - More than one device sharing the link simultaneously

- Time shared Connection:
  - The devices share the link on turn by turn basis

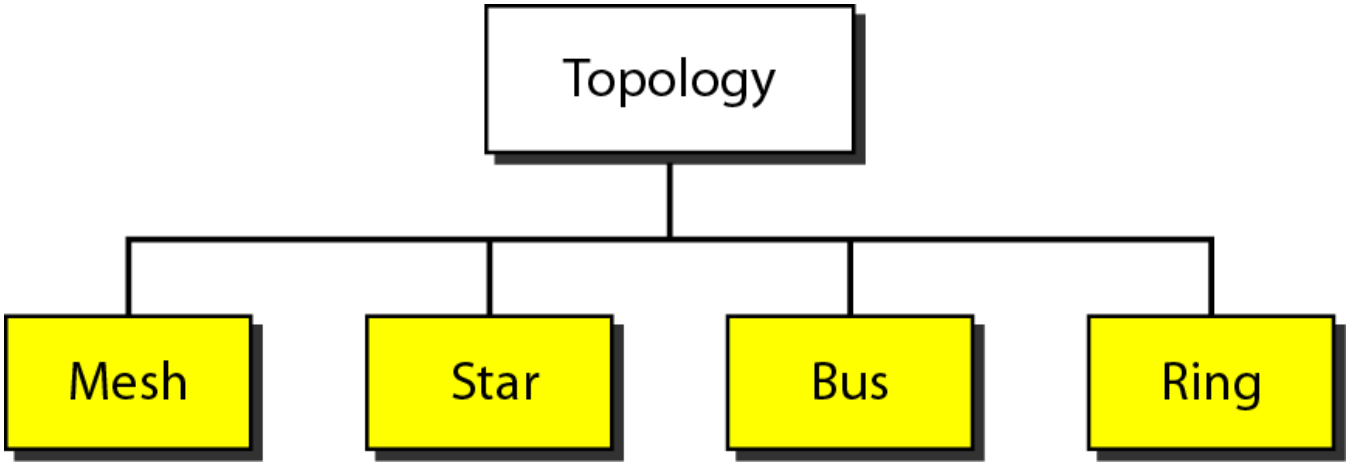
Physical topology

- It refers to the way in which a network is laid out physically.
- Two or more devices connect to a link
- Two or more links form a topology

Topology:

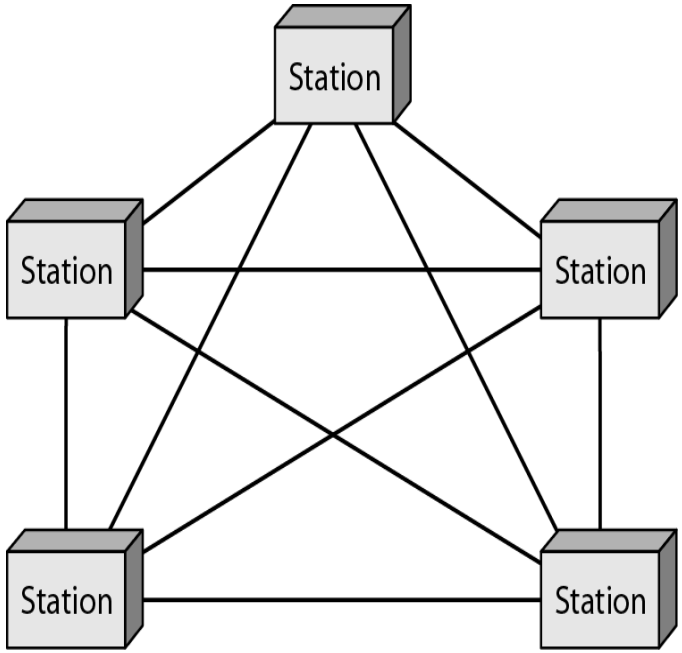
- It is the geometric representation of the relationship of all the links and linking devices(called nodes) to one another

Categories of topology



Mesh Topology

- Every device has a dedicated point-to-point link to every other device.
- To find the number of physical links in a fully connected mesh network with *n* nodes, we first consider that each node must be connected to every other node.
- Node 1 must be connected to *n* - 1 nodes, node 2 must be connected to *n* – 1 nodes, and finally node *n* must be connected to *n* - 1 nodes. We need ***n(n - 1)* physical links**.



- if each physical link allows communication in both directions (duplex mode, we need

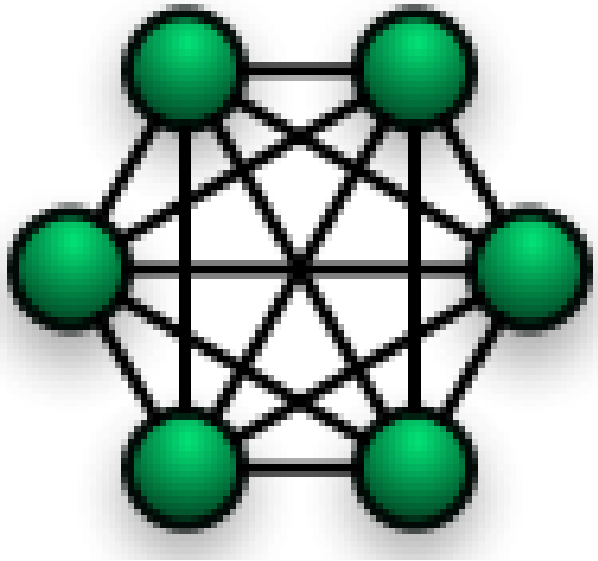
$$n(n-1)/2$$

- Mesh
  - Example:

    6 devices in mesh has links: n(n-1) / 2

    number of links = 6 (6-1)/2 = **15**

    number of ports per device = n – 1 = 6 –1 = 5



Advantages of Mesh topology

- Eliminating the traffic problems
- Robust (If one link becomes unusable, it does not incapacitate the entire system)
- privacy or security
- Fault identification and fault isolation

Disadvantages of Mesh Topology

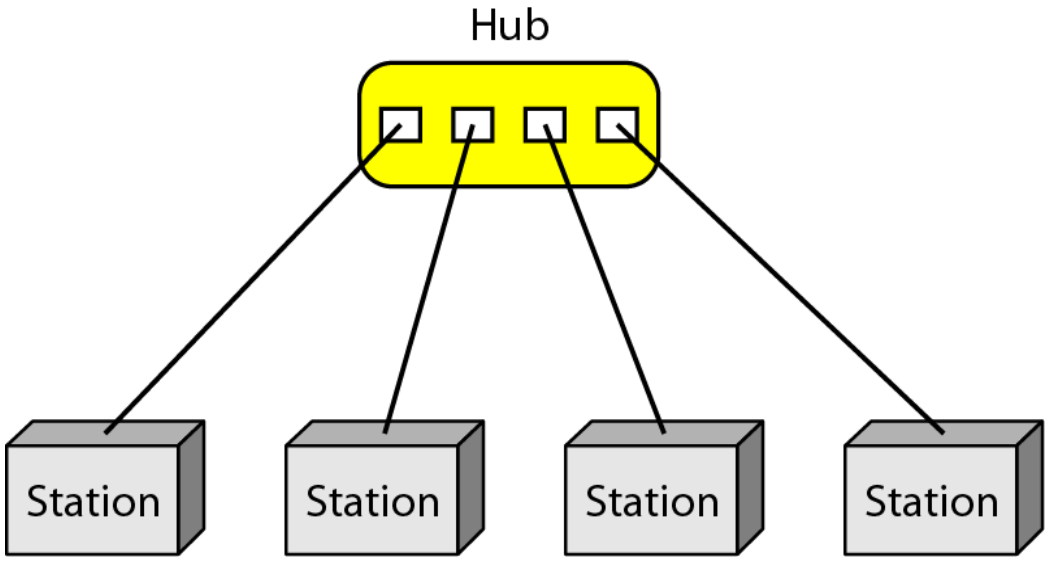
- amount of cabling & number of I/O ports
- installation and reconnection are difficult
- Use of wires(in walls, ceilings, or floors)
- Hardware((I/O ports and cable) can be prohibitively expensive

practical example of a mesh topology

the connection of telephone regional offices in which each regional office needs to be connected to every other regional office

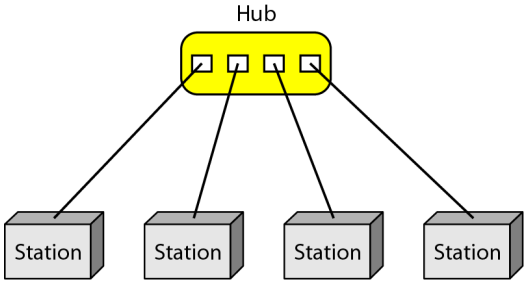
Star Topology

- each device has a dedicated point-to-point link only to a central controller, usually called a **hub**
- The devices are not directly linked to one another
- No direct traffic between devices
- The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device



Advantages

- less expensive than a mesh topology (1 Link + 1 port per device)
- easy to install and reconfigure
- less cabling needs to be housed
- additions, moves, and deletions involve only one connection: between that device and the hub
- Robustnesss(If one link fails, only that link is affected. All other links remain active)



Disadvantage

- dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead

star topology is used in local-area networks (LANs)

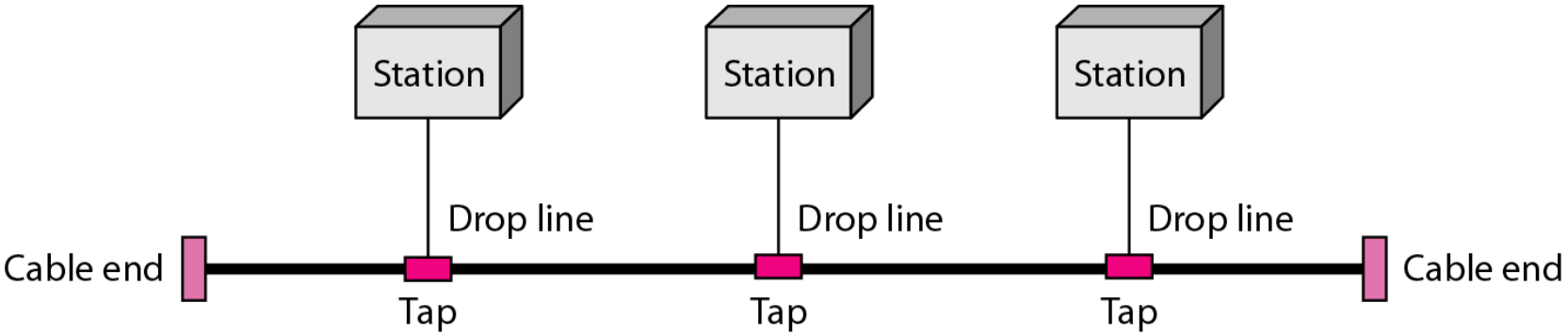


Bus Topology

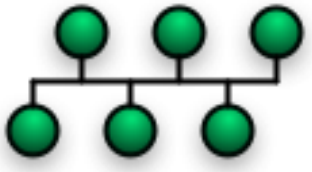
- It is a multipoint connection
- One long cable acts as a backbone
- Used in the design of early LANS, and Ethernet LANs



- Nodes are connected to the bus cable by drop lines and taps.
- A **drop line** is a connection running between the device and the main cable.
- A **tap** is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
- As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther.
- For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.

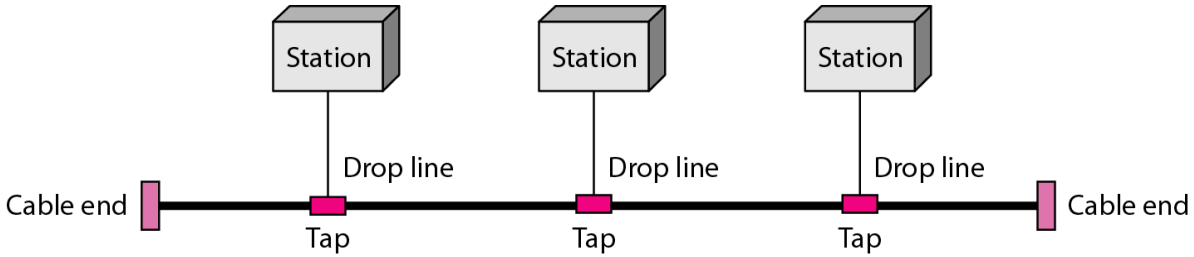


- Bus



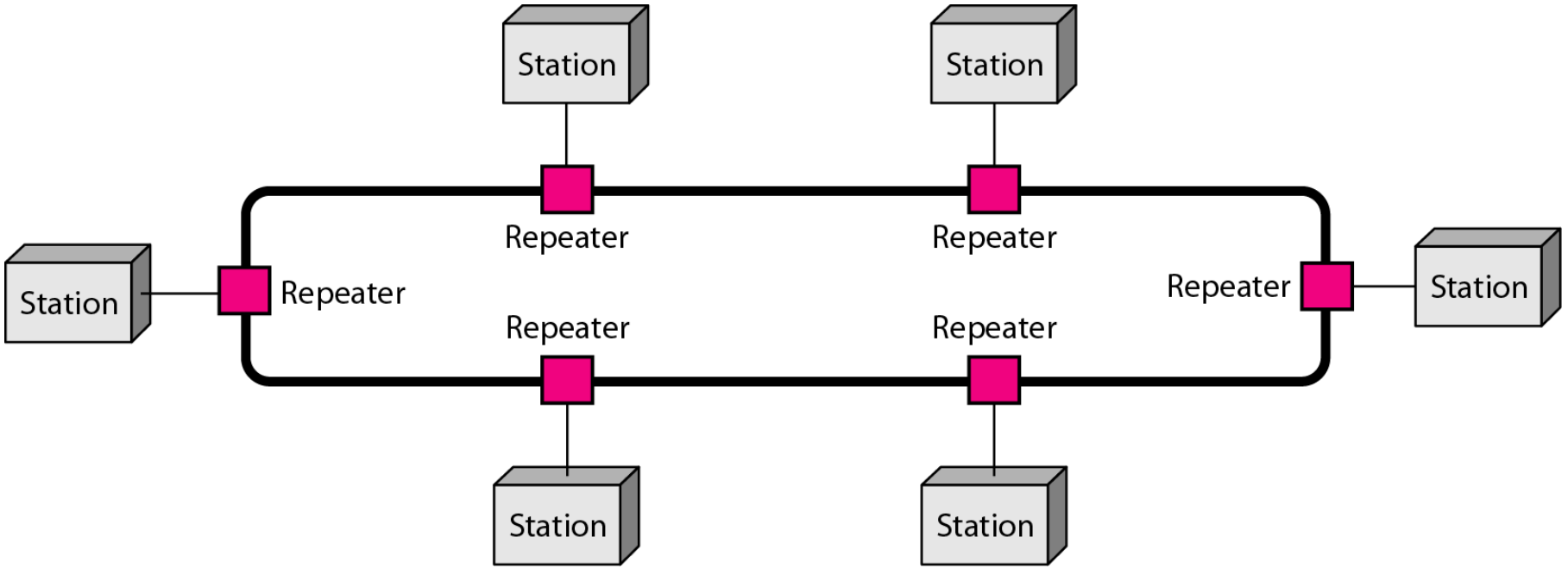
  - Advantages
    - Ease of installation
    - Less cables than mesh, star topologies

  - Disadvantages
    - Difficult reconnection and fault isolation (limit of taps)
    - Adding new device requires modification of backbone
    - Fault or break stops all transmission
    - The damaged area reflects signals back in the direction of the origin, creating noise in both directions

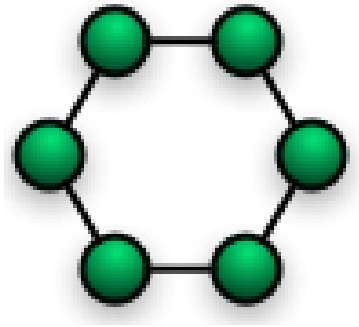


Ring Topology

- Each device has dedicated point-to-point connection with only the two devices on either side of it
- A signal is passed along the ring in one direction from device to device until it reaches its destination
- Each devices incorporates a Repeater



- **Ring**



  - Advantages
    - Easy of install and reconfigure
    - Connect to immediate neighbors
    - Move two connections for any moving (Add/Delete)
    - Easy of fault isolation

  - Disadvantage
    - Unidirectional
    - One broken device can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break

Categories of Networks

Categories of Networks

- we are generally referring to two primary categories:
  - local-area networks and
  - wide-area networks.

- The category into which a network falls is determined by its size.

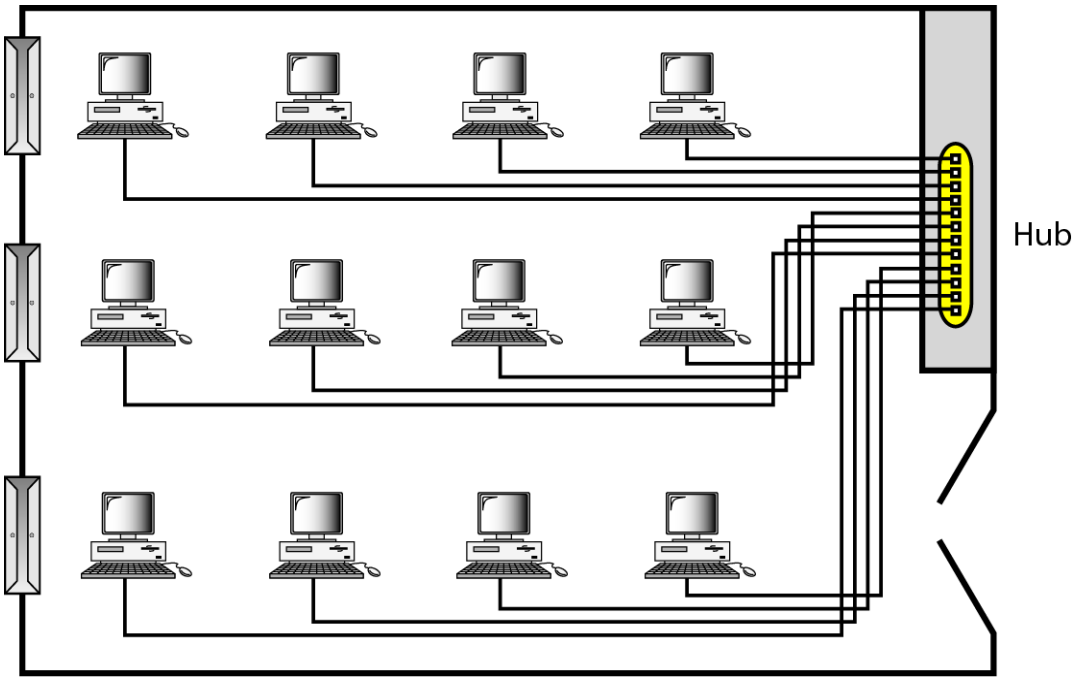
- A **Local Area Network(LAN)** normally covers an area less than 2 miles;

- A **Wide Area Network (WAN)** can be worldwide.

- Networks of a size in between are normally referred to as **Metropolitan Area Networks (MAN)** and span tens of miles.

Local Area Network

- A local area network (LAN) is usually privately owned and links the devices in a single office, building, or campus
- Depending on the needs of an organization and the type of technology used, a LAN can be as simple as two PCs and a printer in someone's home office;
- Currently, LAN size is limited to a few kilometers.



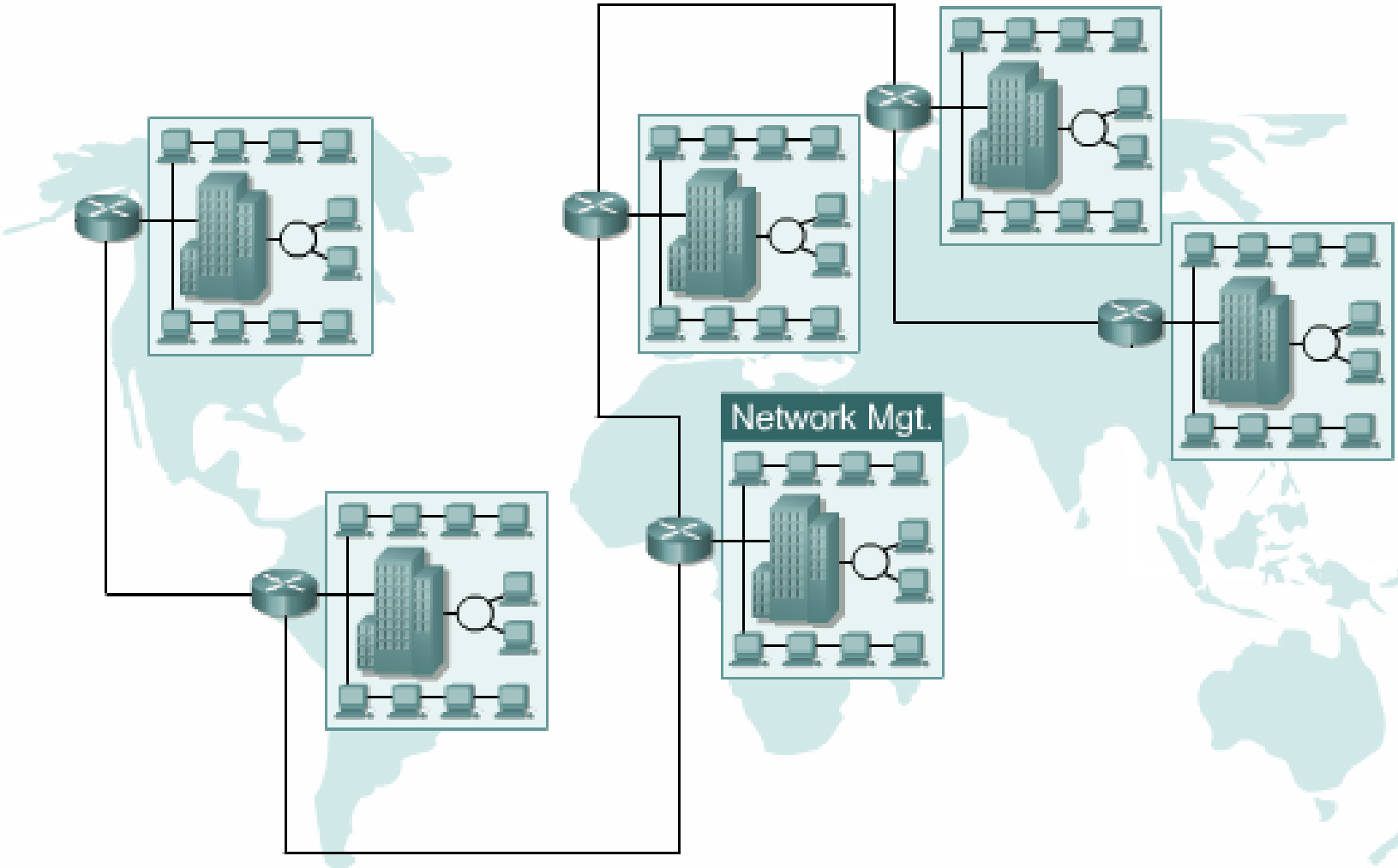
- LANs are designed to allow resources to be shared between personal Computer and Workstations
- LAN will use only one type of transmission medium
- The most common LAN topologies are bus, ring, and star.
- Early LANs had data rates in the 4 to 16 Megabits per second (Mbps) range.
- Today, however, speeds are normally 100 or 1000 Mbps.
- Wireless LANs are the newest evolution in LAN technology.

Wide Area Network

- A wide area network (WAN) provides long-distance transmission of data, image, audio, and video information over large geographic areas that may comprise a country, a continent, or even the whole world.

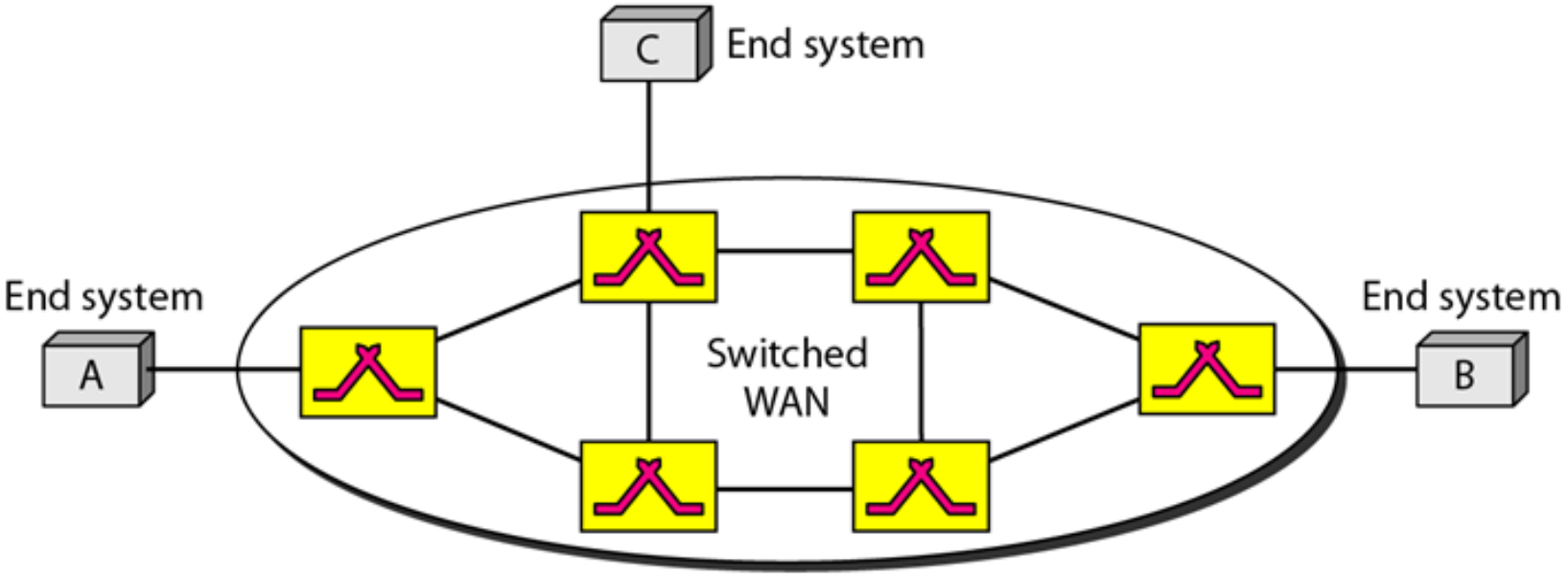
- A WAN can be as complex as the backbones that connect the Internet or as simple as a dial-up line that connects a home computer to the Internet.

- Two types of WAN
  - switched WAN
  - point-to-point WAN


Network Mgt.

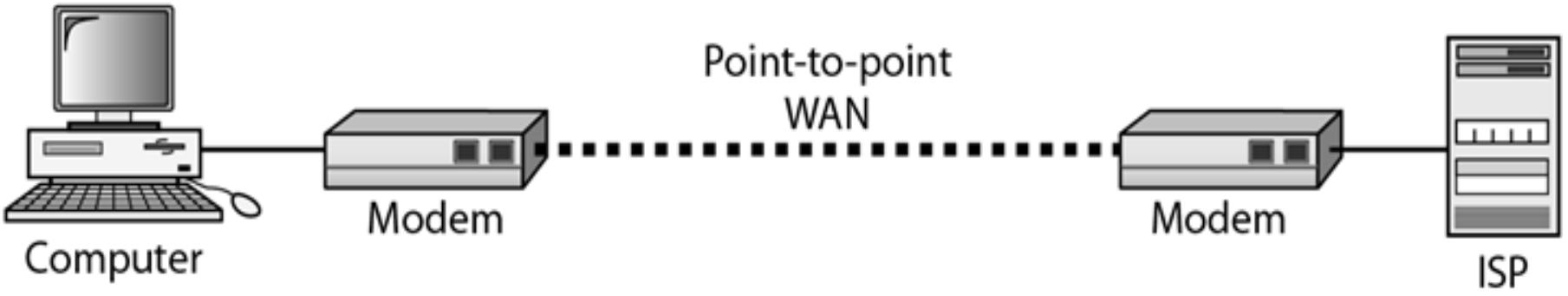
switched WAN

- The switched WAN connects the end systems, which usually comprise a router (internetworking connecting device) that connects to another LAN or WAN.



point-to-point WAN

- The point-to-point WAN is normally a line leased from a telephone or cable TV provider that connects a home computer or a small LAN to an Internet service provider (lSP).
- This type of WAN is often used to provide Internet access.



Examples:

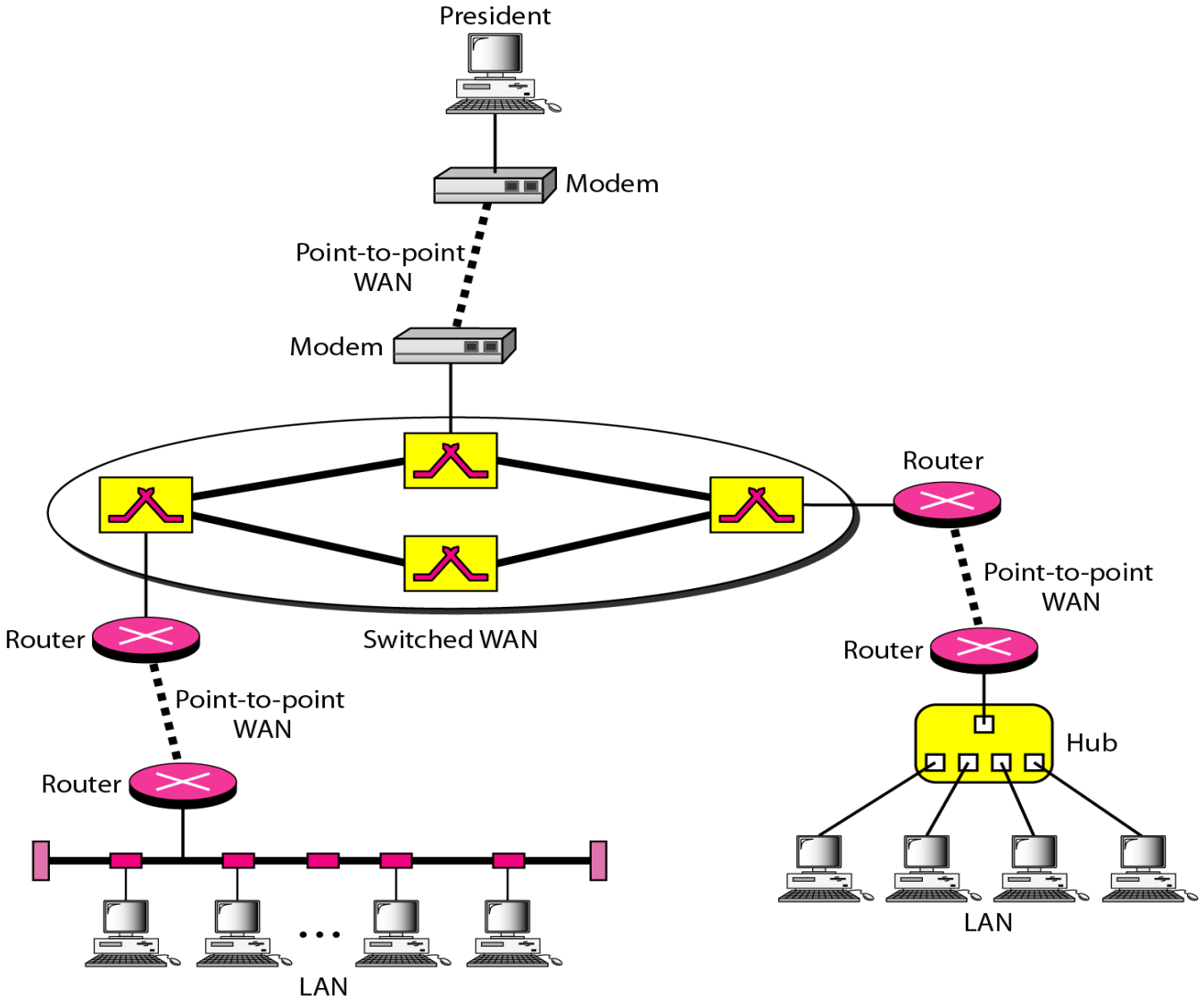
- An early example of switched WAN X.25. It is replaced by high speed network – Frame Relay
- Asynchronous Transfer Mode (ATM)- it is a network with fixed size data unit called cells. Wireless WAN

Metropolitan Area Networks

- A metropolitan area network (MAN) is a network with a size between a LAN and a WAN.
- It normally covers the area inside a town or a city.
- It is designed for customers who need a high-speed connectivity, normally to the Internet, and have endpoints spread over a city or part of city.
- A **good example** of a MAN is the part of the telephone company network that can provide a high-speed DSL line to the customer.
- Another example is the cable TV network that originally was designed for cable TV, but today can also be used for high-speed data connection to the Internet

Internetwork: Interconnection of Networks

A heterogeneous network made of four WANs and two LANs



Protocols and Standards

- Protocol synonymous with rule
- Standards: agreed-upon rules

- Protocols
  - A protocol is a set of rules that govern data communications
  - Defines What, How, and When it is communicated

Protocols and Standards

- Elements of a protocol:
  - ***Syntax:*** structure or format of data
    - Example: 8-bits address of sender, 8-bits address of receiver
  - ***Semantics:*** meaning of each section of bits
    - Example: Does the address is a route to be taken or the final destination of the message
  - ***Timing:*** when data should be sent and how fast they can be sent
    - Example: sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps ⇒ overload and data loose

Standards

- Essential in creating and maintaining an open and competitive market for equipment manufactures
- Guaranteeing national and international interoperability of data and telecommunication technology and processes
- Providing guidelines to manufacturers, vendors, government agencies, and other service providers to ensure the kind of interconnectivity necessary in today's marketplace and in international communications

Standards

- Data Communication standards fall into two categories
  - De facto (by fact or by convention):
    - not approved by an organized body but adopted as standards through widespread use.
  - De jure (by law or by regulation):
    - Legislated by an officially recognized body

Standards

- Standards are developed through the cooperation of:
  - Standards Creation Committees
    - ISO, ITU-T, CCITT, ANSI, IEEE, EIA
  - Forums
    - Created by special-interest groups
    - Present their conclusions to the standards bodies
  - Regulatory Agencies
    - Ministry of Telecommunication and Information Technology (KSA)
    - Purpose: Protecting the public by regulating radio, television, and communication

Network Models

- A network is a combination of hardware and software that sends data from one location to another.

- The ***hardware*** consists of the physical equipment that carries signals from one point of the network to another.

- The ***software*** consists of instruction sets that make possible the services that we expect from a network.

Network Models

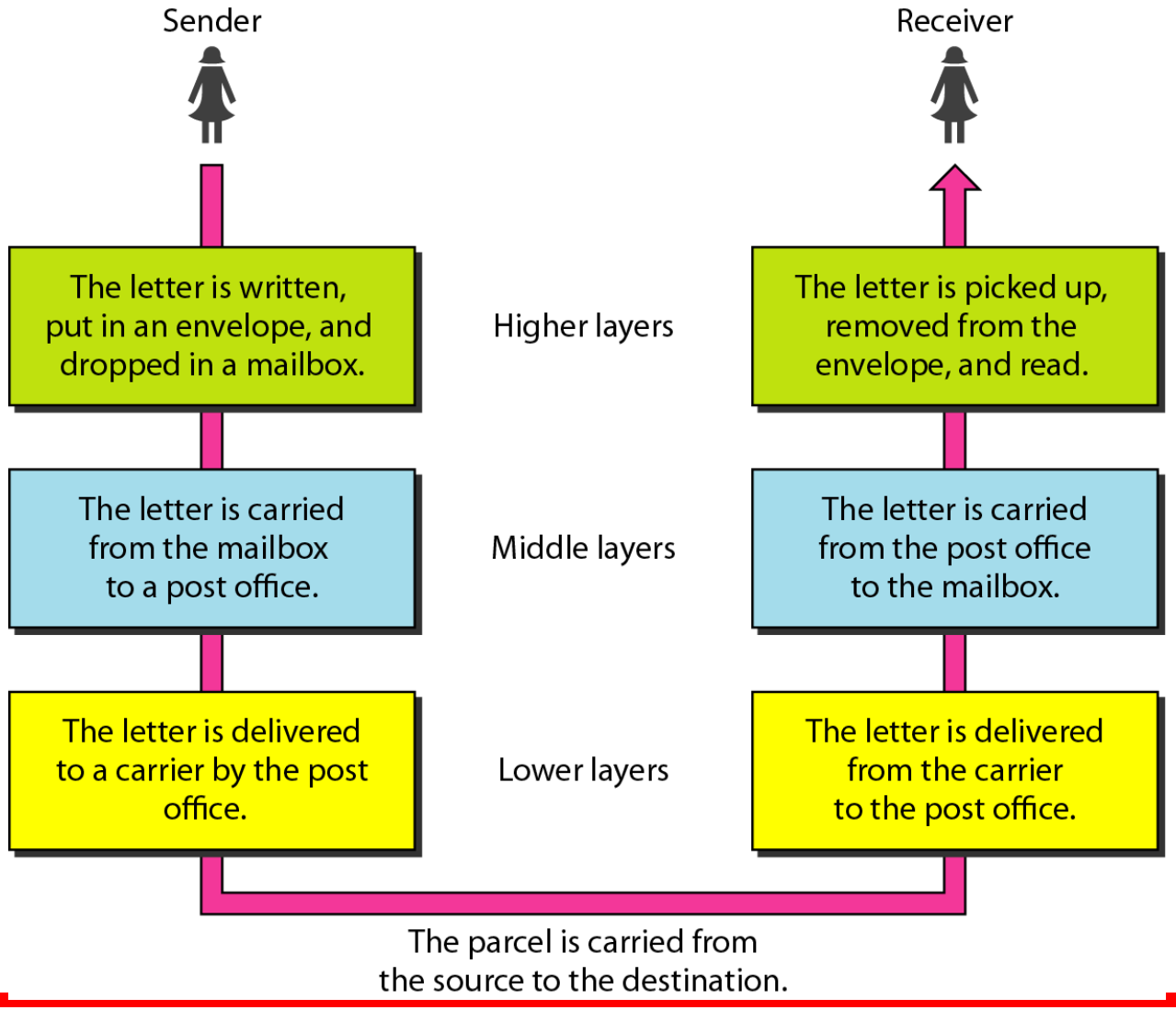
- Computer networks are created by different entities.
- Standards are needed so that these heterogeneous networks can communicate with one another.
- The two best-known models are
  - OSI model and the Internet model.
- The **OSI (Open Systems Interconnection)** model defines a seven-layer network;
- the **Internet model** defines a five-layer network.

LAYERED TASKS

- We use the concept of layers in our daily life.

- As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.

Tasks involved in sending a letter



Hierarchy

- According to our analysis, there are three different activities at the sender site and another three activities at the receiver site.

- The task of transporting the letter between the sender and the receiver is done by the carrier.

- Something that is not obvious immediately is that the tasks must be done in the order given in the hierarchy.

- At **the sender site**, the letter must be written and dropped in the mailbox before being picked up by the letter carrier and delivered to the post office.
- At **the receiver site**, the letter must be dropped in the recipient mailbox before being picked up and read by the recipient.

Network Architecture

- A computer network must provide general, cost effective and robust connectivity among a large number of computers.

- To deal with this complexity network designers developed the concept of network architecture which introduces a layered protocols or functions that guides the design and implementation of networks

Most widely network architectures are

- OSI Architecture
- Internet Architecture

Why Layering?

1) It decomposes the problem of building a network into more manageable components. Each layer solves one part of problem

2) It provides a more modular design. That is, if we want to add new service, it requires only modifying the functionality at the respective layers and reusing the functions provided at all the other layers

Services

- Each layer at the sending site uses the services of the layer immediately below it.
- The sender at the higher layer uses the services of the middle layer. The middle layer uses the services of the lower layer.
- The lower layer uses the services of the carrier.
- The layered model that dominated data communications and networking literature before 1990 was the Open Systems Interconnection (OSI) model.
- Everyone believed that the OSI model would become the ultimate standard for data communications, but this did not happen.

THE OSI MODEL

- Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards.

- An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model.

- It was first introduced in the late 1970s.

- An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.

- The **purpose of the OSI model** is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software.

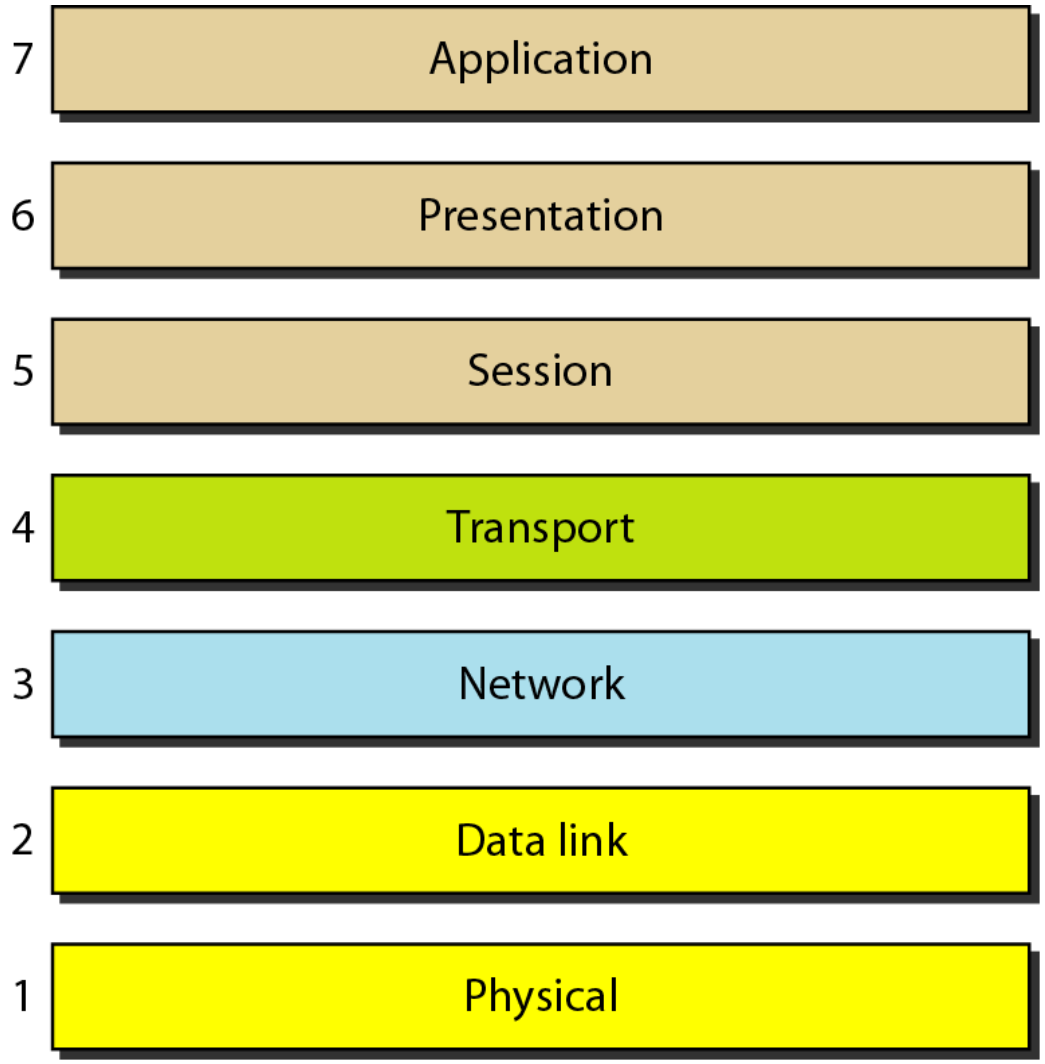
- The OSI model is not a protocol;

- It is a model for understanding and designing a network architecture that is flexible, robust, and interoperable.

Note

ISO is the organization.
OSI is the model.

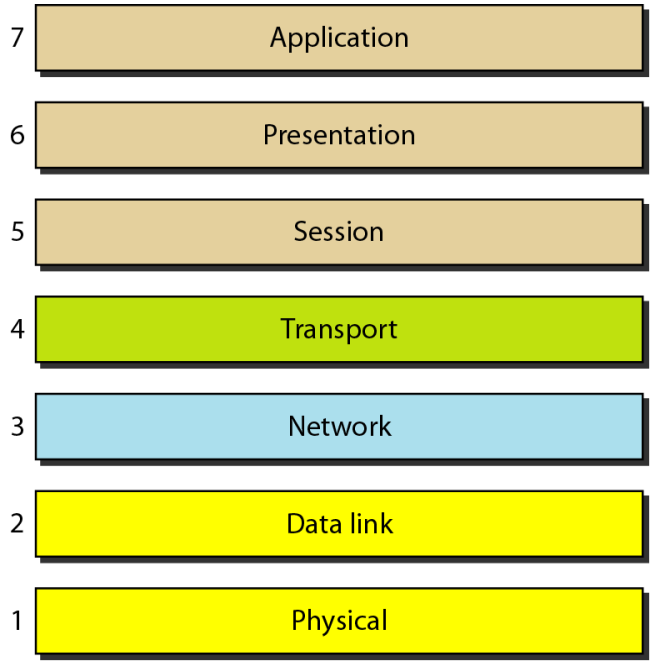
Seven layers of the OSI model



Seven layers can be split into three subgroups

Layer 1,2,3

- Network support layers
- It deals with physical aspects of moving data from one device to another (electrical specifications, physical connection, physical addressing, timing and reliability)



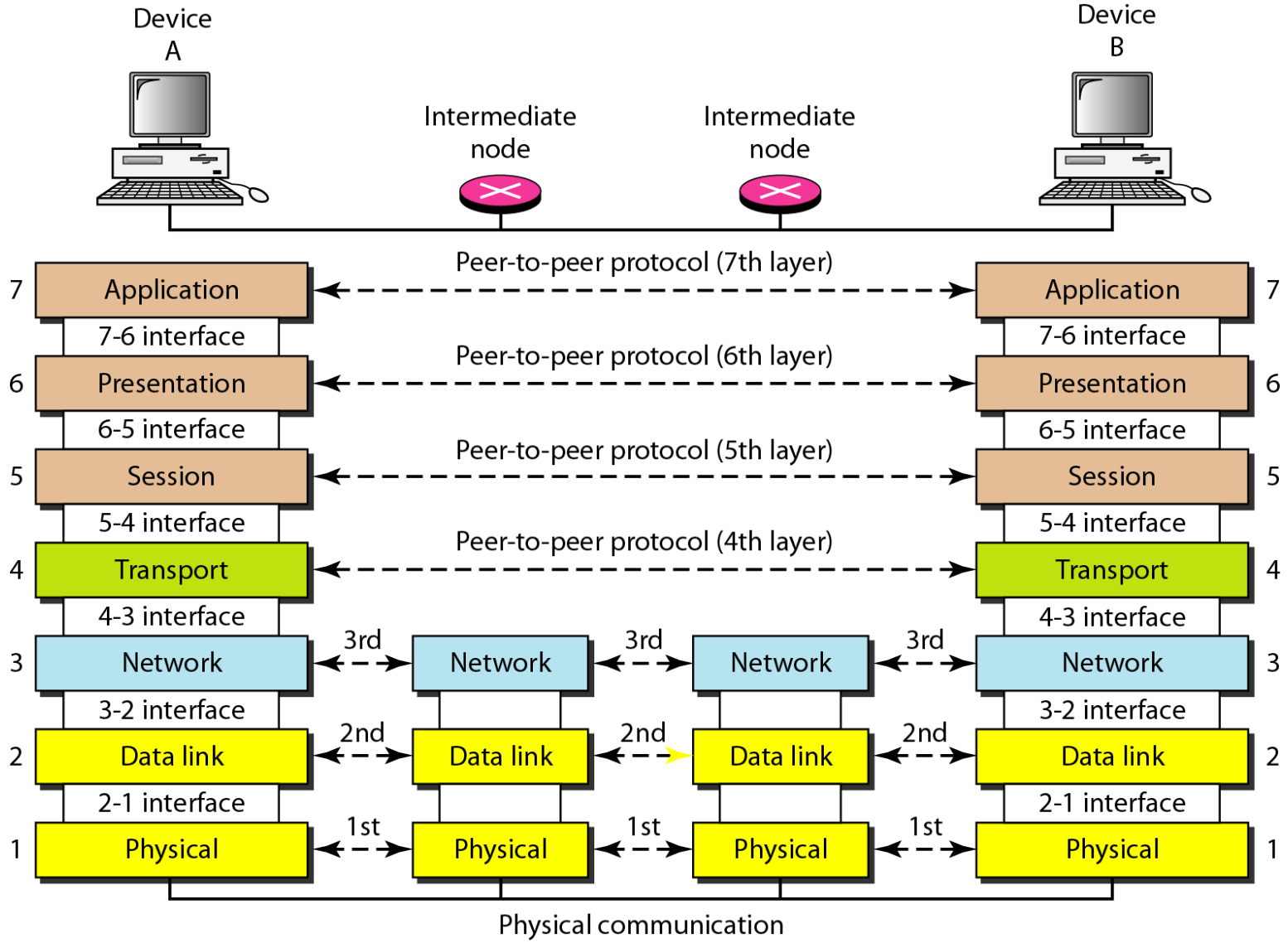
Layer 5,6&7 – (implemented in software)

- User support layers
- It allows interoperability among unrelated software systems

Layer 4

- It links the two subgroups (lower layers and upper layer)
- It ensures end-to-end reliable data transmission

The interaction between layers in the OSI model



- Between machines, layer *x* on one machine communicates with layer *x* on another machine.

- This communication is governed by an agreed-upon series of rules and conventions called protocols.

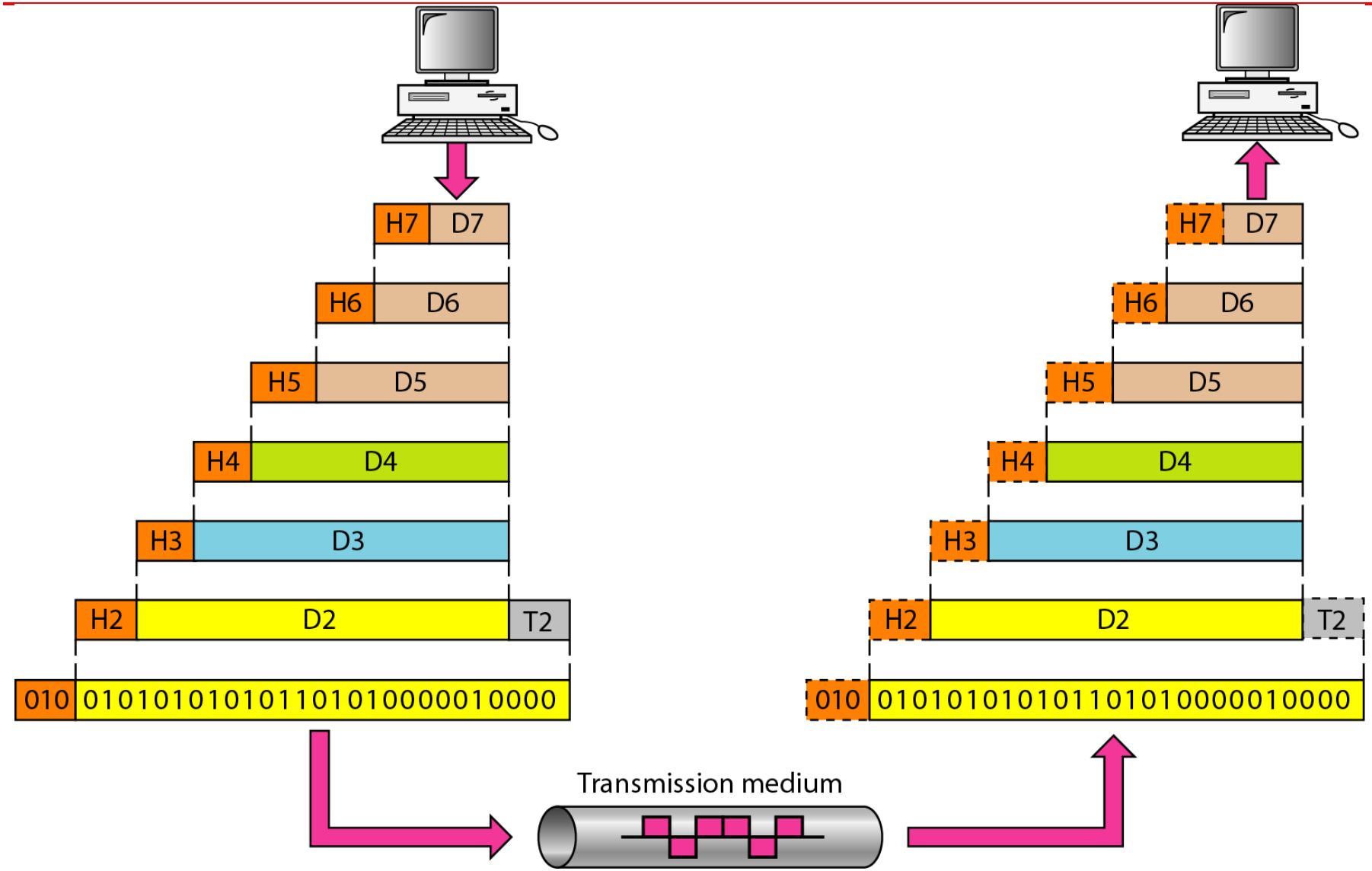
- The processes on each machine that communicate at a given layer are called peer-to-peer processes.

- Communication between machines is therefore a peer-to-peer process using the protocols appropriate to a given layer

Peer-to-Peer Processes

- At the physical layer, communication is direct:
- device A sends a stream of bits to device B (through intermediate nodes).
- At the higher layers, however, communication must move down through the layers on device A, over to device B, and then back up through the layers.
- Each layer in the sending device adds its own information to the message it receives from the layer just above it and passes the whole package to the layer just below it.

An exchange using the OSI model



- D7 means the data unit at layer 7, D6 means the data unit at layer 6, and so on.

- The process starts at layer 7 (the application layer), then moves from layer to layer in descending, sequential order.

- At each layer, a **header,** or possibly a **trailer,** can be added to the data unit.

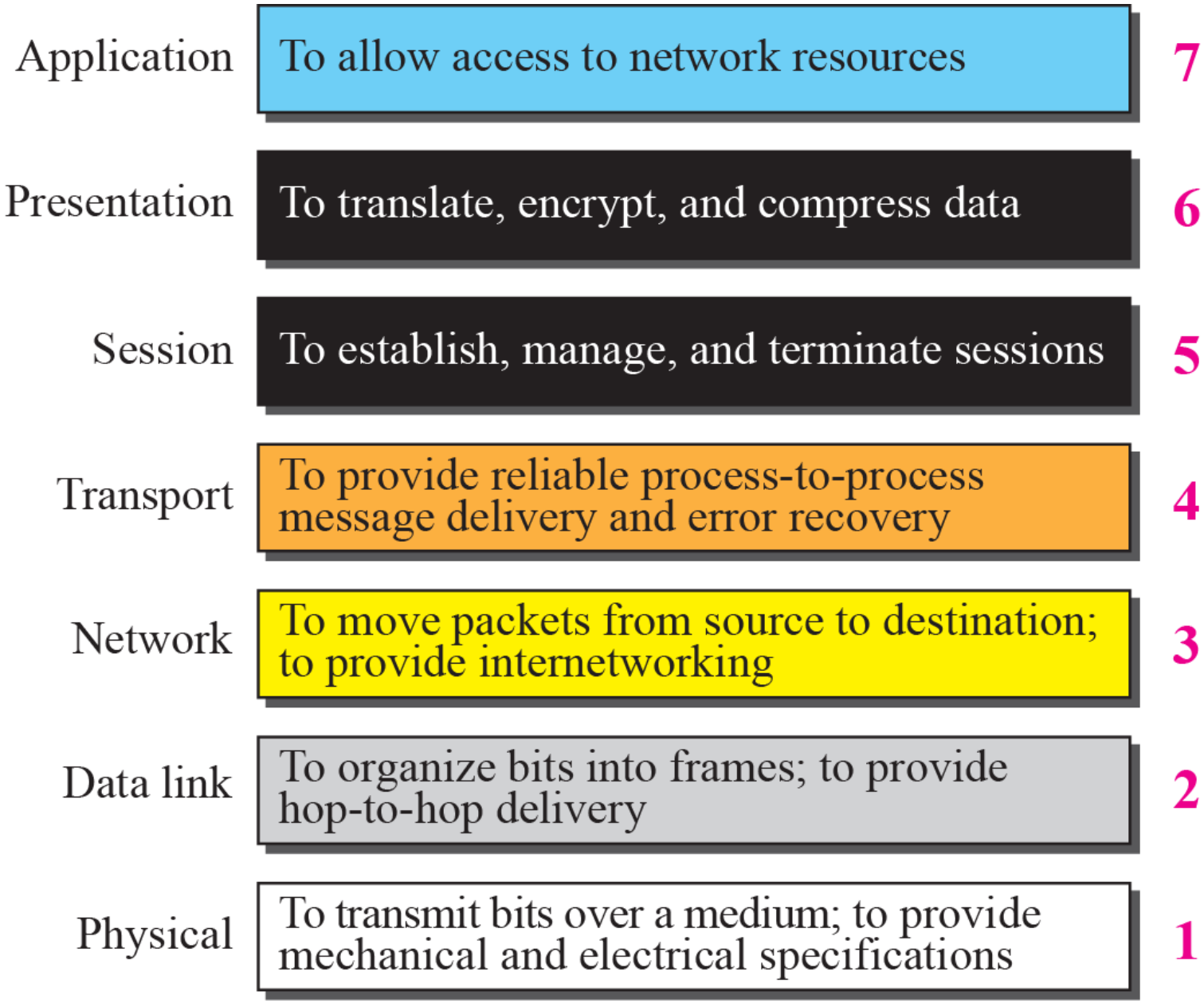
- Commonly, the trailer is added only at layer 2.

- When the formatted data unit passes through the physical layer (layer 1), it is changed into an electromagnetic signal and transported along a physical link.

Encapsulation

- Another aspect of data communications in the OSI model: encapsulation.
- A packet (header and data) at level 7 is encapsulated in a packet at level 6.
- The whole packet at level 6 is encapsulated in a packet at level 5, and so on.
- In other words, the data portion of a packet at level *N* - 1 carries the whole packet (data and header and maybe trailer) from level *N*. The concept is called *encapsulation;*
- level *N* - 1 is not aware of which part of the encapsulated packet is data and which part is the header or trailer.
- For level *N* - 1, the whole packet coming from level *N* is treated as one integral unit.

Summary of OSI Layers

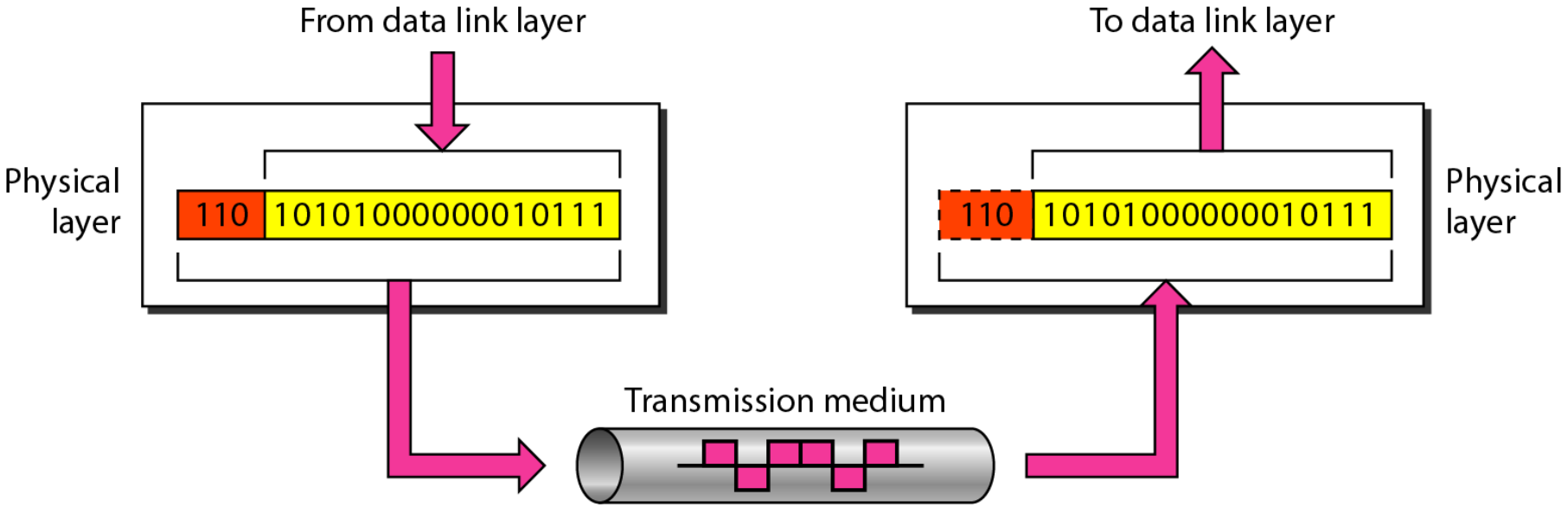


OSI Layers

LAYERS IN THE OSI MODEL

Physical Layer

- The physical layer coordinates the functions required to carry a bit stream over a physical medium.
- It deals with the mechanical and electrical specifications of the interface and transmission medium.
- It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to Occur.



Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

The physical layer is also concerned with the following:

- **Physical characteristics of interfaces and medium.**
  - The physical layer defines the characteristics of the interface between the devices and the transmission medium.
  - It also defines the type of transmission medium.
- **Representation of bits.**
  - data consists of a stream of bits (sequence of Os or 1s) with no interpretation.
  - To be transmitted, bits must be encoded into signals-electrical or optical.
  - It defines the type of encoding (how Os and I s are changed to signals).

- **Data rate.**
  - The ***transmission rate***-the number of bits sent each second-is also defined by the physical layer.

- **Synchronization of bits.**
  - The sender and receiver not only must use the same bit rate.
  - The sender and the receiver clocks must be synchronized.

- **Line configuration.**
  - The physical layer is concerned with the connection of devices to the media.
  - In a point-to-point configuration, two devices are connected through a dedicated link.
  - In a multipoint configuration, a link is shared among several devices.

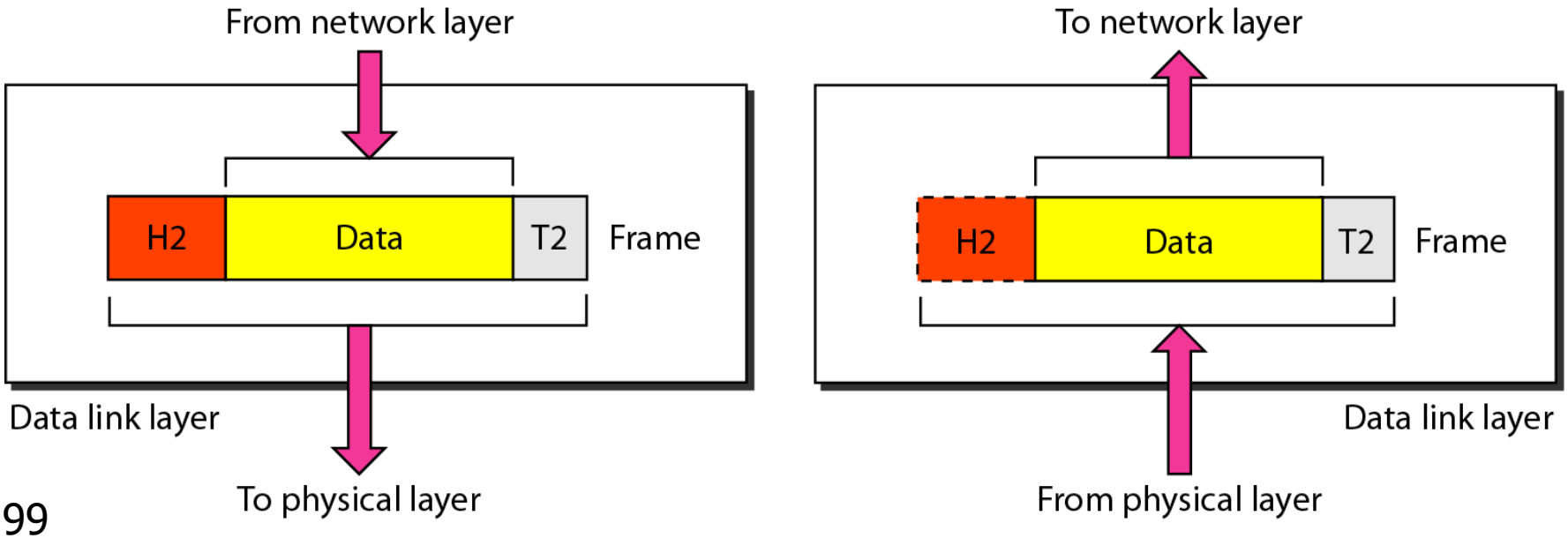
- **Physical topology.**
  - The physical topology defines how devices are connected to make a network.
  - Devices can be connected by using
    - a mesh topology
    - a star topology
    - a ring topology a
    - a bus topology
    - a hybrid topology

- **Transmission mode.**
  - The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex.

Data Link Layer

- It makes the physical layer appear error-free to the upper layer (network layer)
- **node – to—node delivery**
  - It is *responsible for transmitting a group of bits* (called frames or packets) to adjacent node without error:

The data link layer is responsible for moving frames from one hop (node) to the next.



Data Link layer

- **Framing -** Divides the stream of bits received into data units called frames.

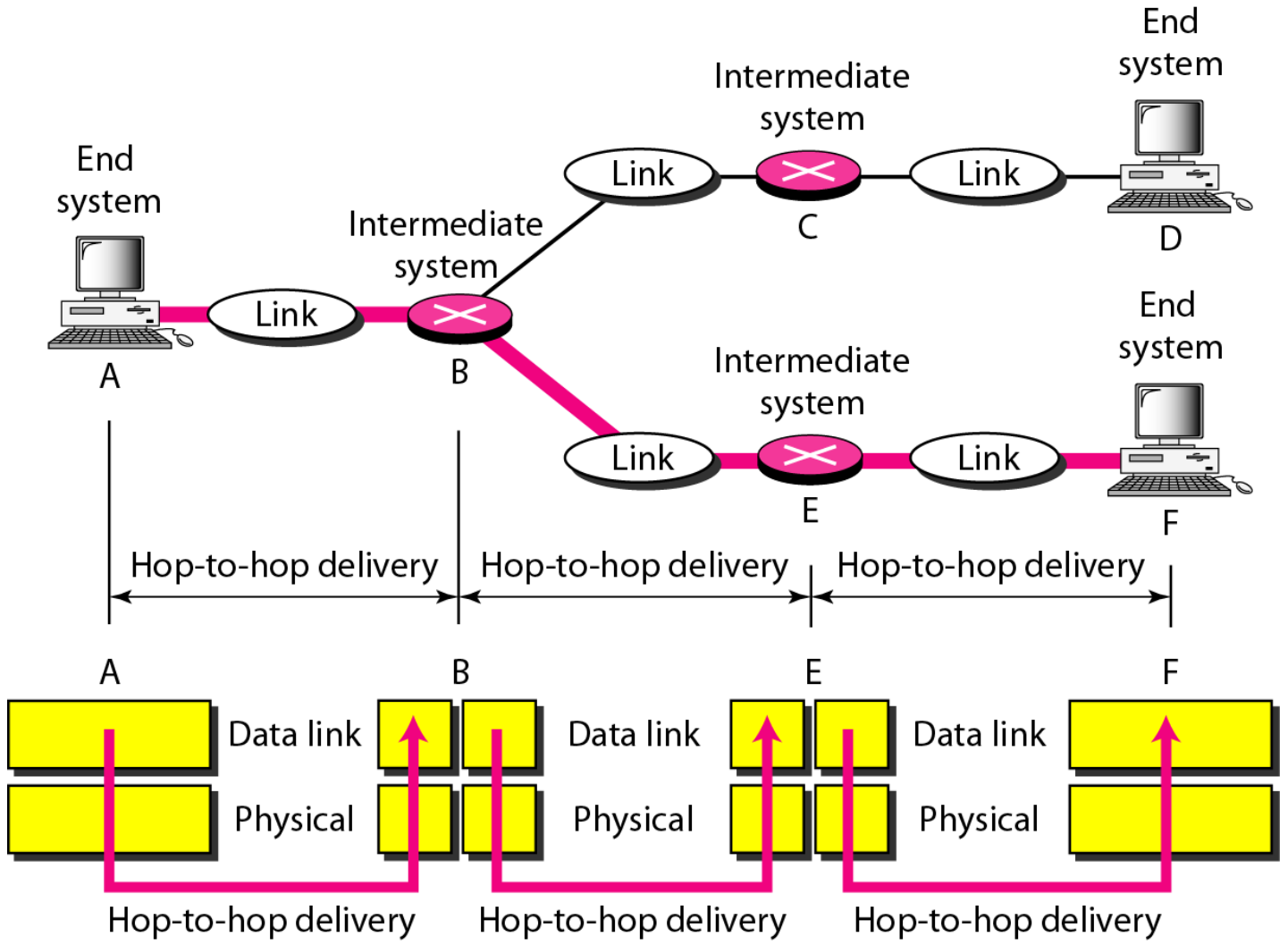
- **Physical addressing** – If frames are to be distributed to different systems on the network, data link layer adds a header to the frame to define the sender and receiver.

- **Flow control -** If the rate at which the data are absorbed by the receiver is less than the rate produced in the sender ,the Data link layer imposes a flow control mechanism.

- **Error control** - Used for detecting and retransmitting damaged or lost frames and to prevent duplication of frames. This is achieved through a trailer added at the end of the frame.

- **Access control** -Used to determine which device has control over the link at any given time when 2 or more devices are connected to the same link.

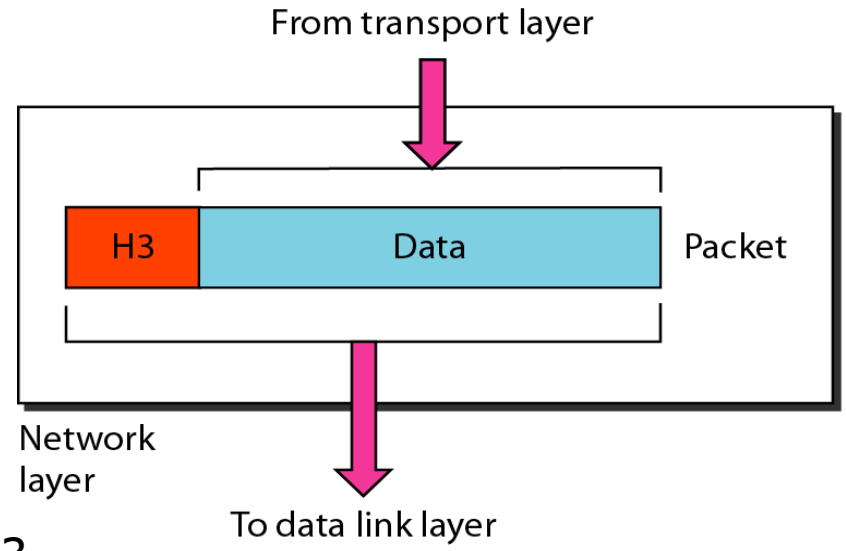
Hop-to-hop delivery

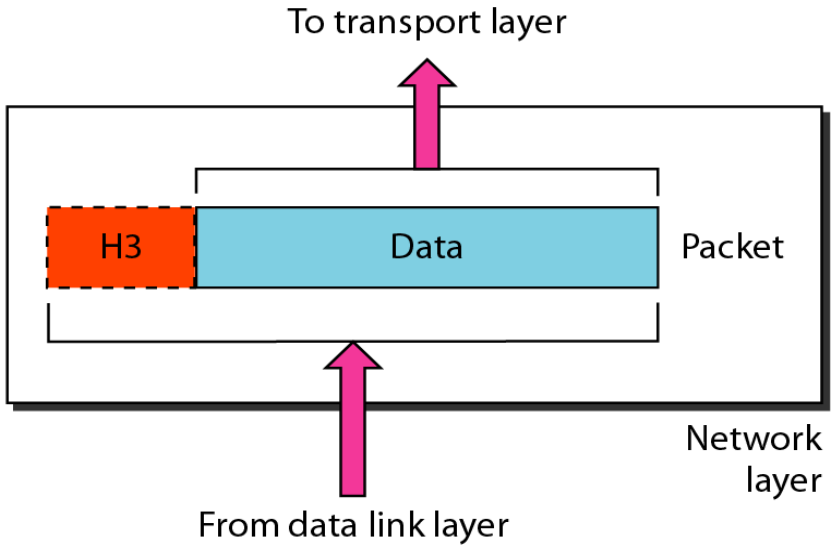


- As the figure shows, communication at the data link layer occurs between two adjacent nodes.
- To send data from A to F, three partial deliveries are made.
- First, the data link layer at A sends a frame to the data link layer at B (a router). Second, the data link layer at B sends a new frame to the data link layer at E. Finally, the data link layer at E sends a new frame to the data link layer at F.
- Note that the frames that are exchanged between the three nodes have different values in the headers.
- The frame from A to B has B as the destination address and A as the source address. Likewise B to E and E to F.
- The values of the trailers can also be different if error checking includes the header of the frame.

Network Layer

- This layer is responsible for the delivery of **packets from source to destination**
- If two systems are connected to the same link, there is usually no need for a network layer.
- However, if the two systems are attached to different networks (links) with connecting devices between the networks (links), there is often a need for the network layer to accomplish source-to-destination delivery.



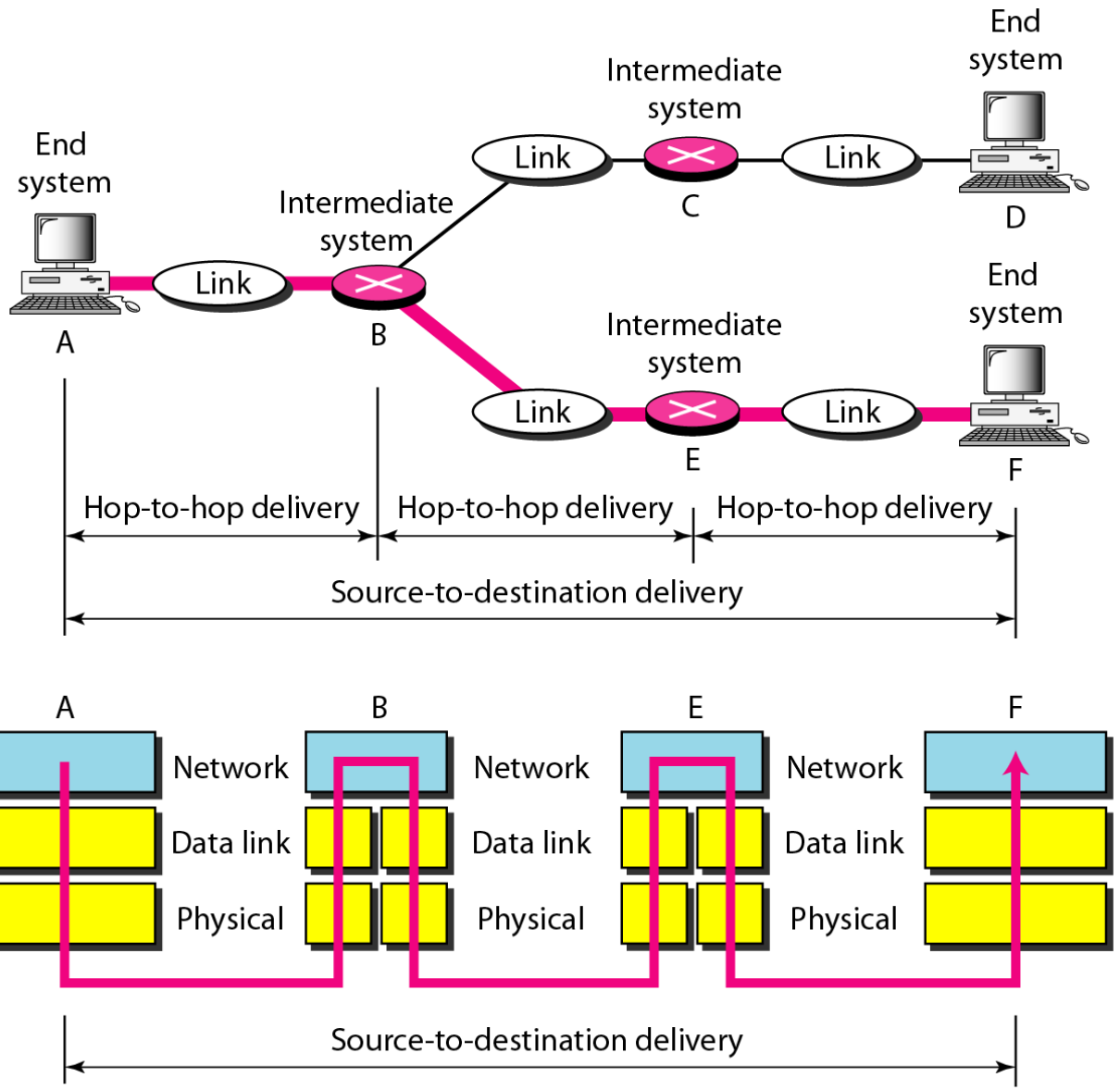


The network layer is responsible for the delivery of individual packets from the source host to the destination host.

- **Logical addressing**
  - If a packet passes the network boundary, we need another addressing system for source and destination called logical address.

- **Routing**
  - The devices which connects various independent networks called routers.
  - Routers are responsible for delivering packets to final destination.

Source-to-destination delivery

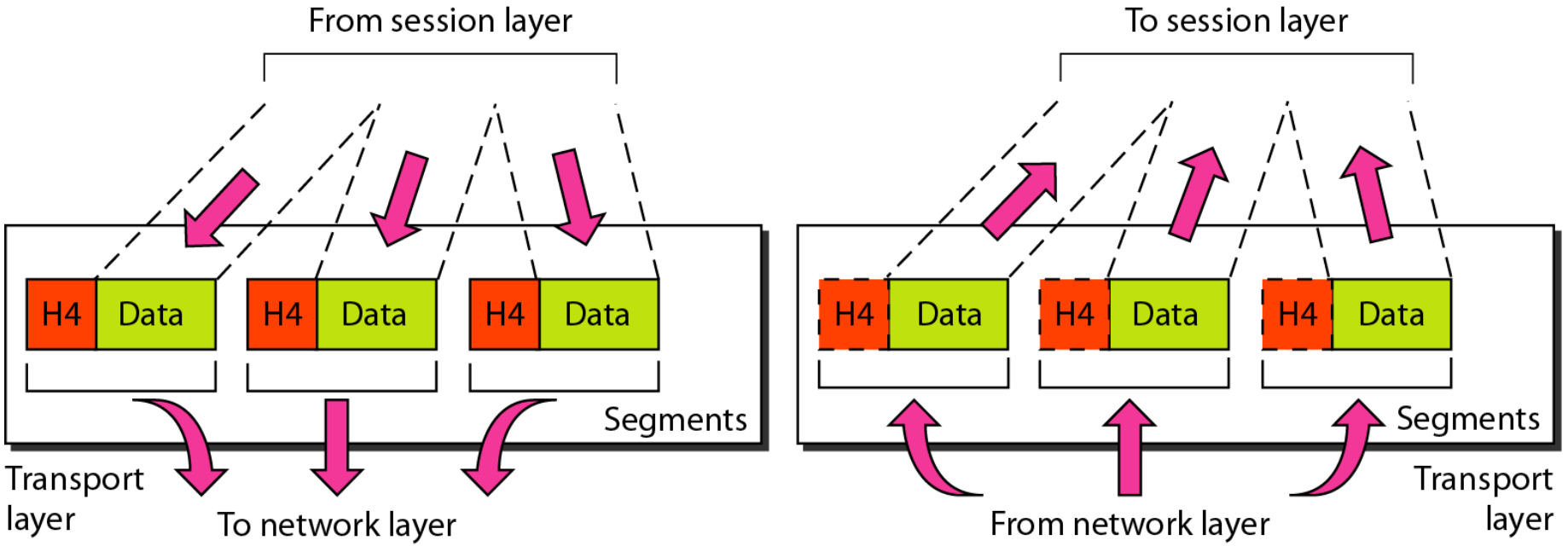


- The network layer at A sends the packet to the network layer at B.
- When the packet arrives at router B, the router makes a decision based on the final destination (F) of the packet.
- router B uses its routing table to find that the next hop is router E.
- The network layer at B, therefore, sends the packet to the network layer at E.
- The network layer at E, in turn, sends the packet to the network layer at F.

Transport Layer

- It is responsible for **process-to-process delivery** of the entire message
- A process is an application program running on a host.
- It does not recognize any relationship between the packets which is arrived from network layer.
- It treats each one independently, as though each piece belonged to a separate message, whether or not it does.
- The transport layer, on the other hand, ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level.

Transport layer



The transport layer is responsible for the delivery of a message from one process to another.

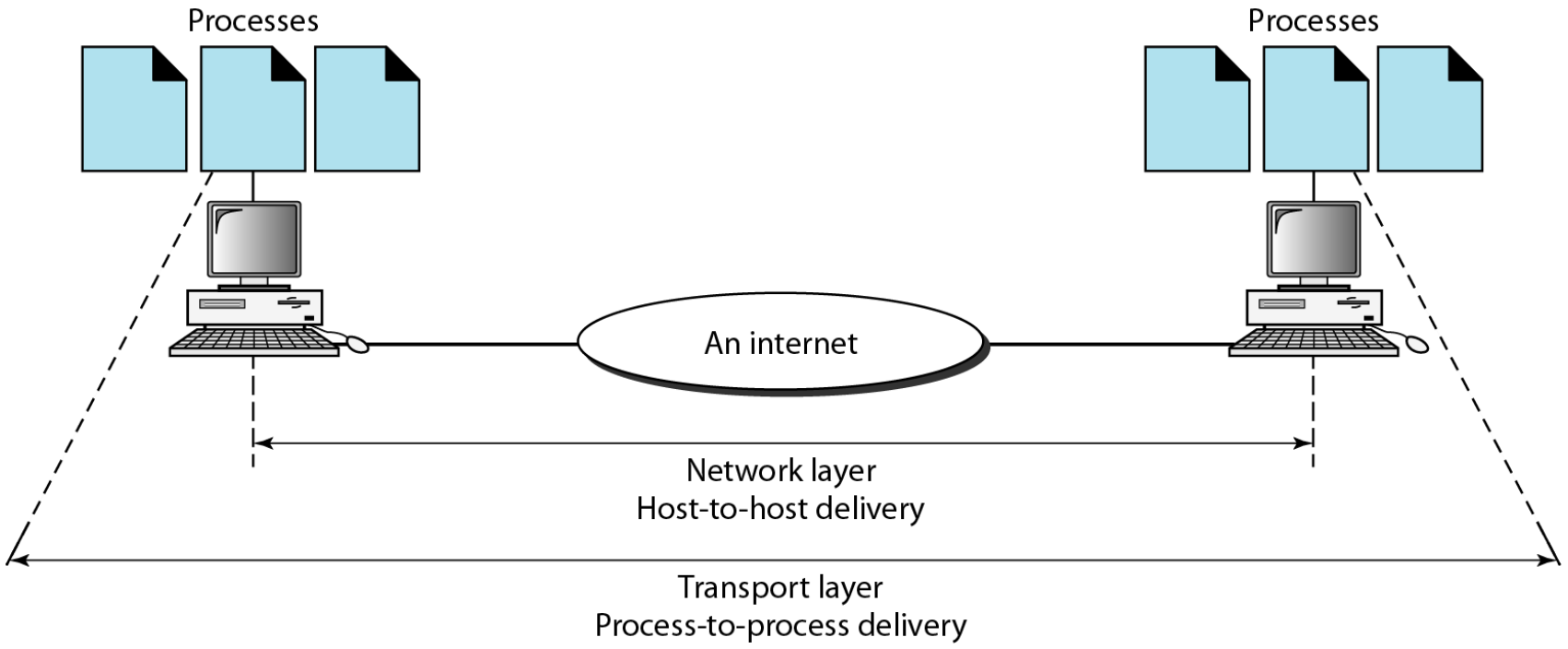
- **Port addressing** - The header in this must therefore include a address called port address or service-point address. This layer gets the entire message to the correct process on that computer.

- **Segmentation and reassembly** - The message is divided into segments and each segment is assigned a sequence number. These numbers are arranged correctly on the arrival side by this layer.

- **Connection control** - This can either be **connectionless or connection-oriented.** The connectionless treats each segment as a individual packet and delivers to the destination. The connection-oriented makes connection on the destination side before the delivery. After the delivery the connection will be terminated.

- **Flow and error control** - Similar to data link layer, but process to process take place.

Reliable process-to-process delivery of a message

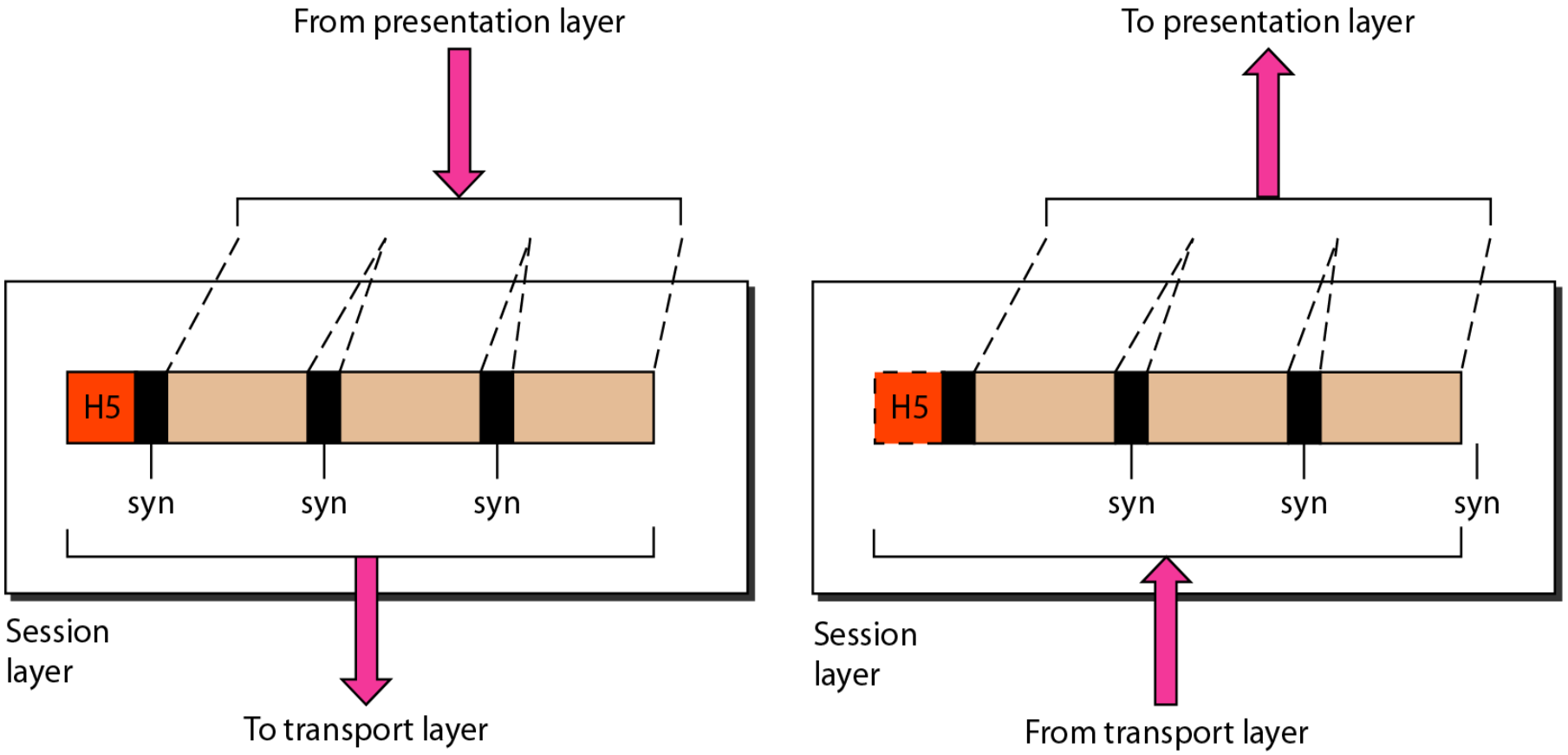


Session Layer

- The services provided by the first three layers (physical, data link, and network) are not sufficient for some processes.
- The session layer is the network *dialog controller*.
- It establishes, maintains, and synchronizes the interaction among communicating systems.
- (i.e) This layer establishes, manages and terminates connections between applications.

The session layer is responsible for dialog control and synchronization.

Session layer



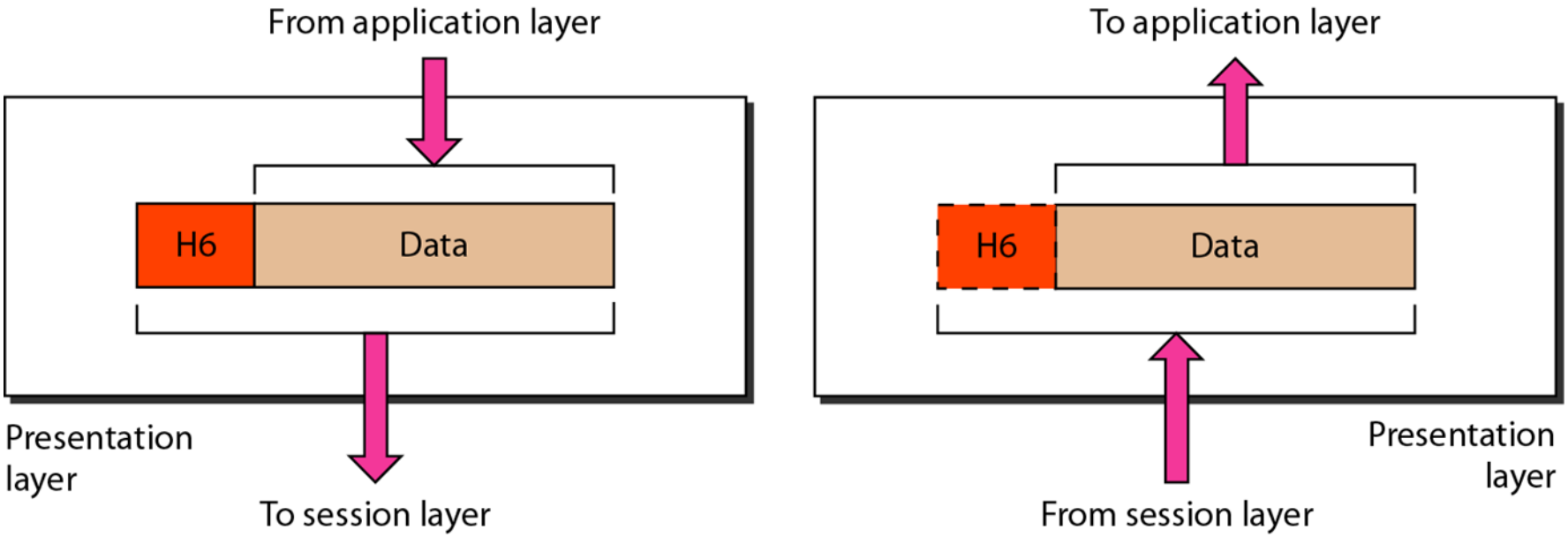
- **Dialog control**
  - This session allows two systems to enter into a dialog either in half duplex or full duplex.

- **Synchronization**
  - This allows to add checkpoints (or synchronization points) into a stream of data.
  - For example, if a system is sending a file of 2000 pages, it is advisable to insert checkpoints after every 100 pages to ensure that each 100-page unit is received and acknowledged independently.
  - In this case, if a crash happens during the transmission of page 523, the only pages that need to be resent after system recovery are pages 501 to 523. Pages previous to 501 need not be resent.

Presentation Layer

- The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems.

The presentation layer is responsible for translation, compression, and encryption.



- **Translation**
  - Different computers use different encoding system, this layer is responsible for interoperability between these different encoding methods.
  - It will change the message into some common format.

- **Encryption and decryption**
  - It means that sender transforms the original information to another form and sends the resulting message over the network and vice versa.

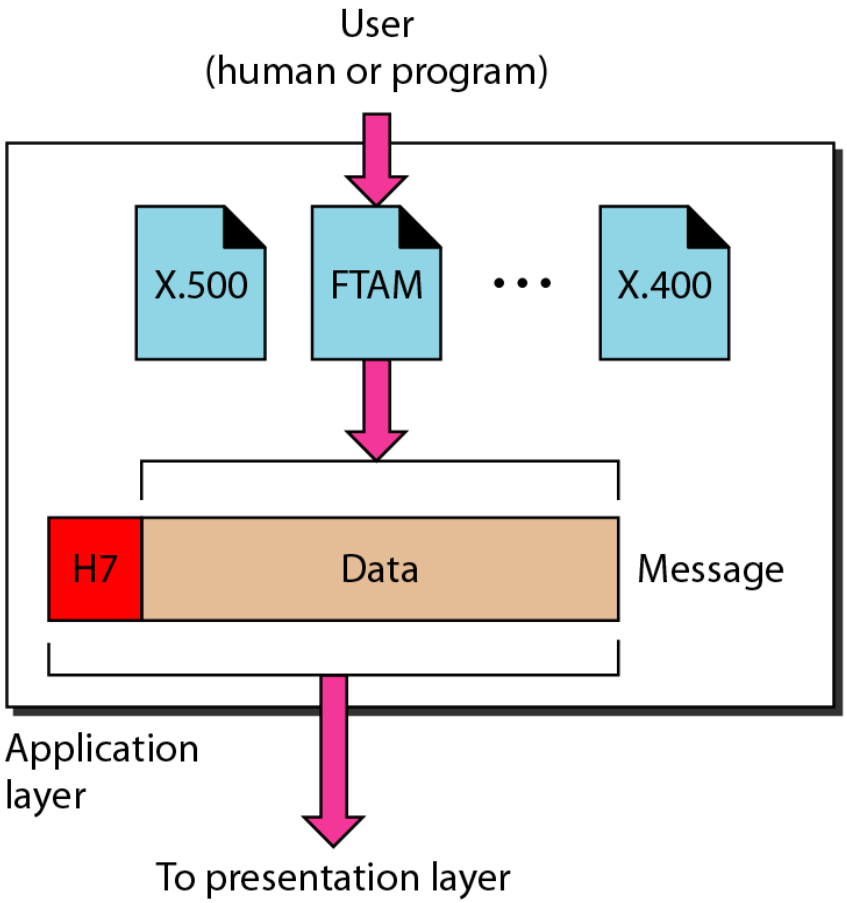
- **Compression and expansion**
  - Compression reduces the number of bits contained in the information particularly in text, audio and video.

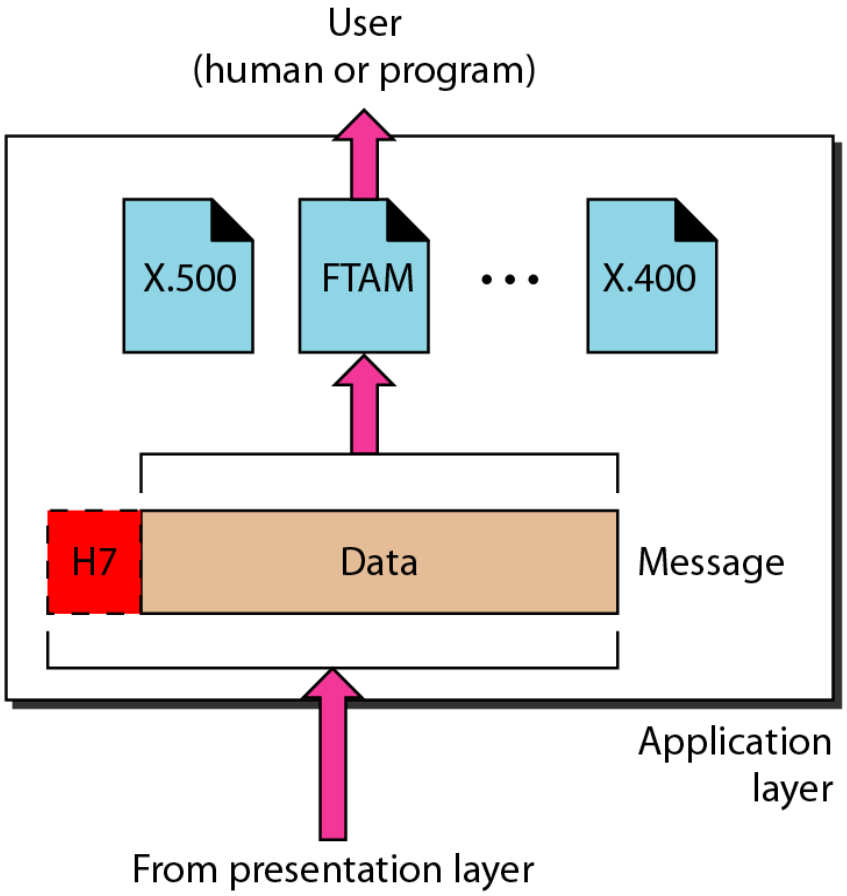
Application Layer

- The application layer enables the user, whether human or software, to access the network.
- It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.
- Application services
  - X.400 (message handling service)
  - X.500 (Directory services)
  - FTAM (file transfer, access and management)

The application layer is responsible for providing services to the user.

Application layer



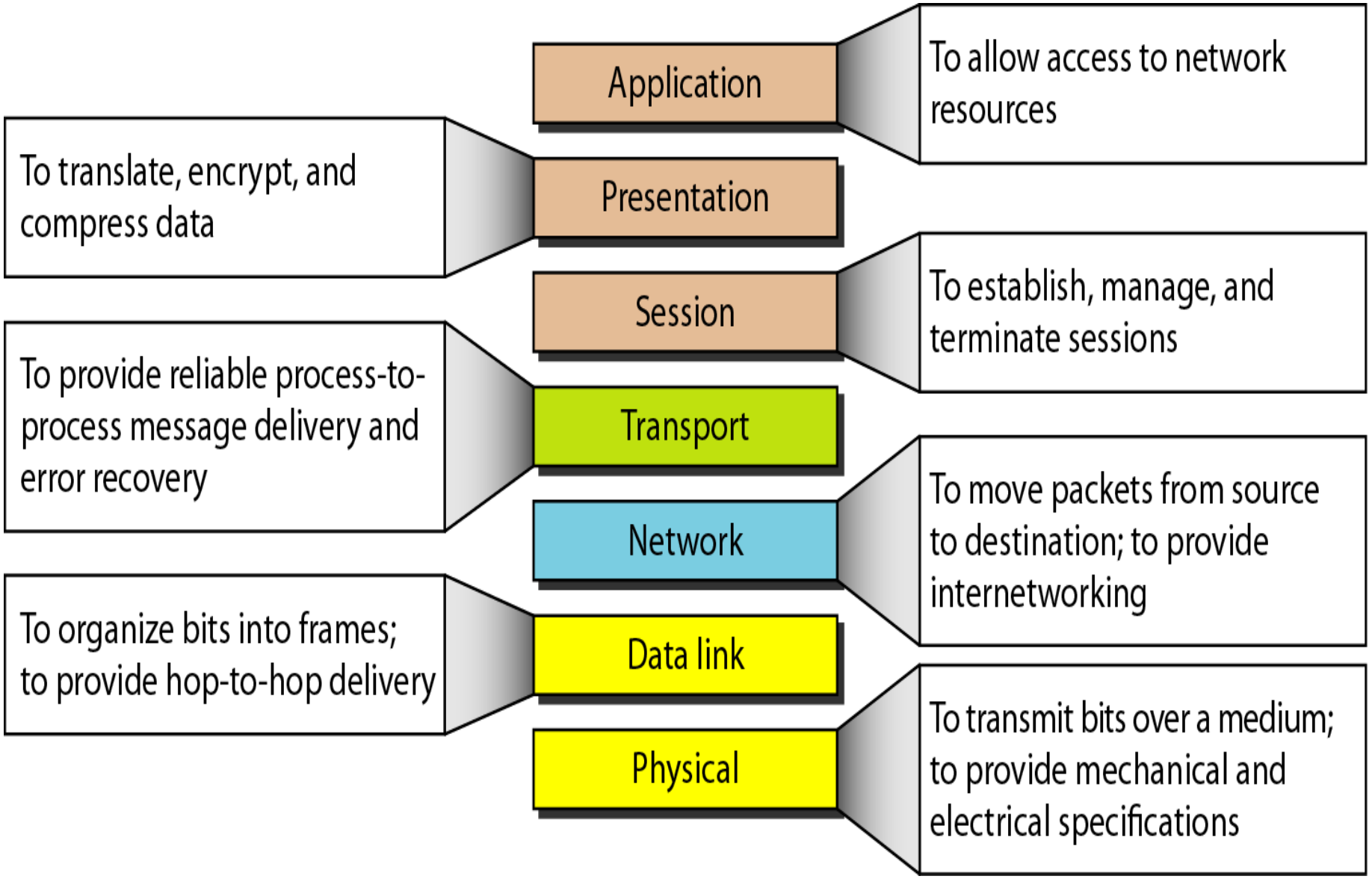


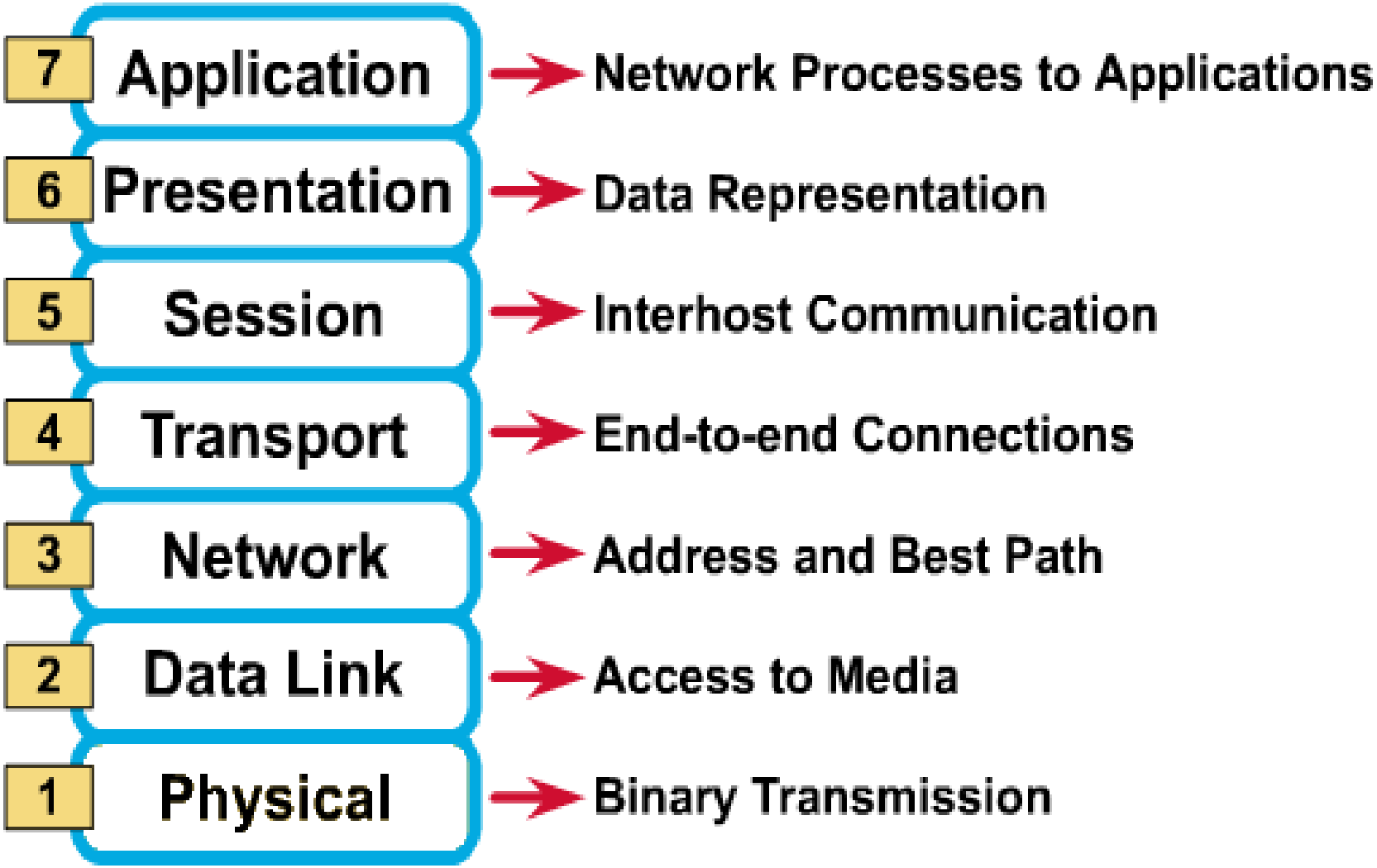
- **FTAM(file transfer, access, mgmt)** - Allows user to access files in a remote host.

- **Mail services** - Provides email forwarding and storage.

- **Directory services** - Provides database sources to access information about various sources and objects.

Summary of layers




7 Application
Network Processes to Applications
6 Presentation
Data Representation
5 Session
Interhost Communication
4 Transport
End-to-end Connections
3 Network
Address and Best Path
2 Data Link
Access to Media
1 Physical
Binary Transmission

Internet Architecture

Internet Architecture

Transmission Control Protocol/ Internetworking Protocol (TCP/IP)

- TCP/IP is a set of rules and procedures, that defines how all transmissions are exchanged across the internet.

- It is also called TCP/IP protocol suite (or) Internet model (or) TCP/IP reference model (or) internet architecture

TCP/IP PROTOCOL SUITE

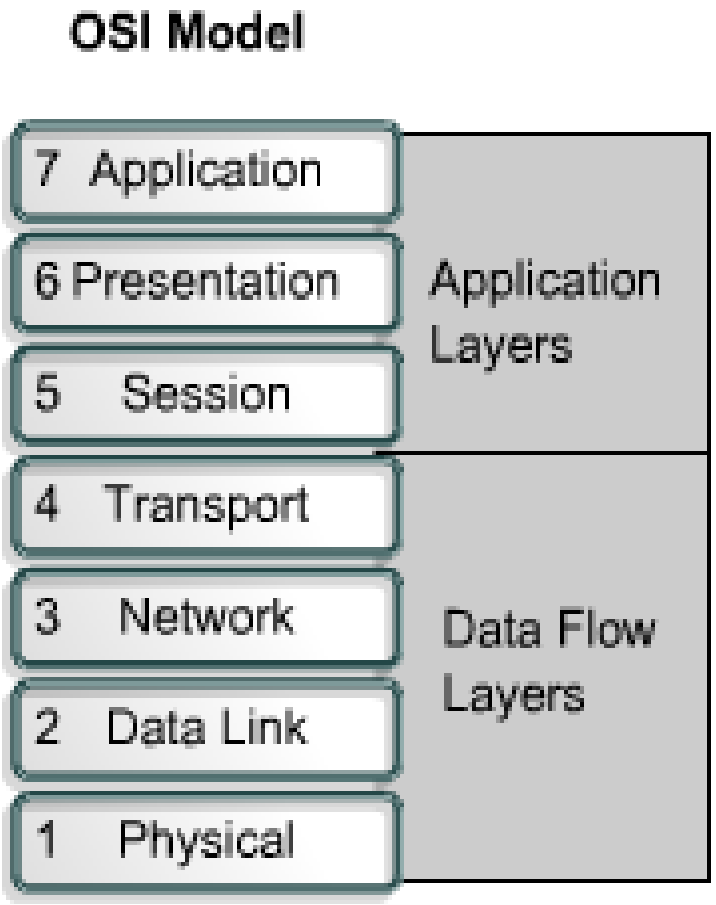
- The TCPIIP protocol suite was developed prior to the OSI model.
- Therefore, the layers in the TCP/IP protocol suite do not exactly match those in the OSI model.
- The original TCP/IP protocol suite was defined as having four layers:
  - host-to-network,
  - internet,
  - transport, and
  - application.
- However, when TCP/IP is compared to OSI, we can say that the host-to-network layer is equivalent to the combination of the physical and data link layers.

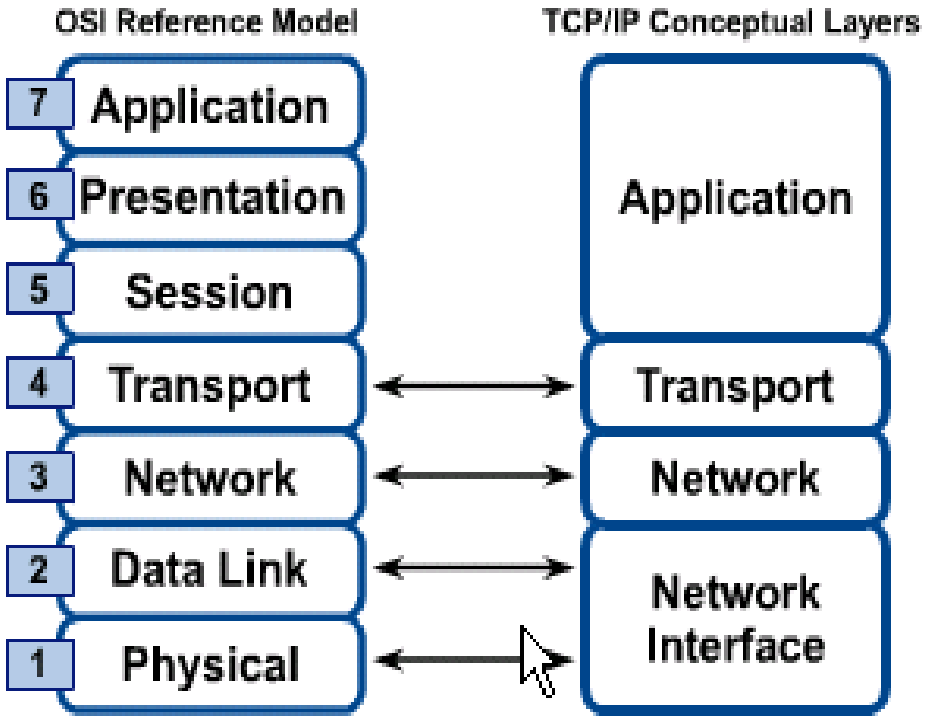
- The internet layer is equivalent to the network layer,

- application layer is roughly doing the job of the session, presentation, and application layers with the transport layer in TCPIIP taking care of part of the duties of the session layer.

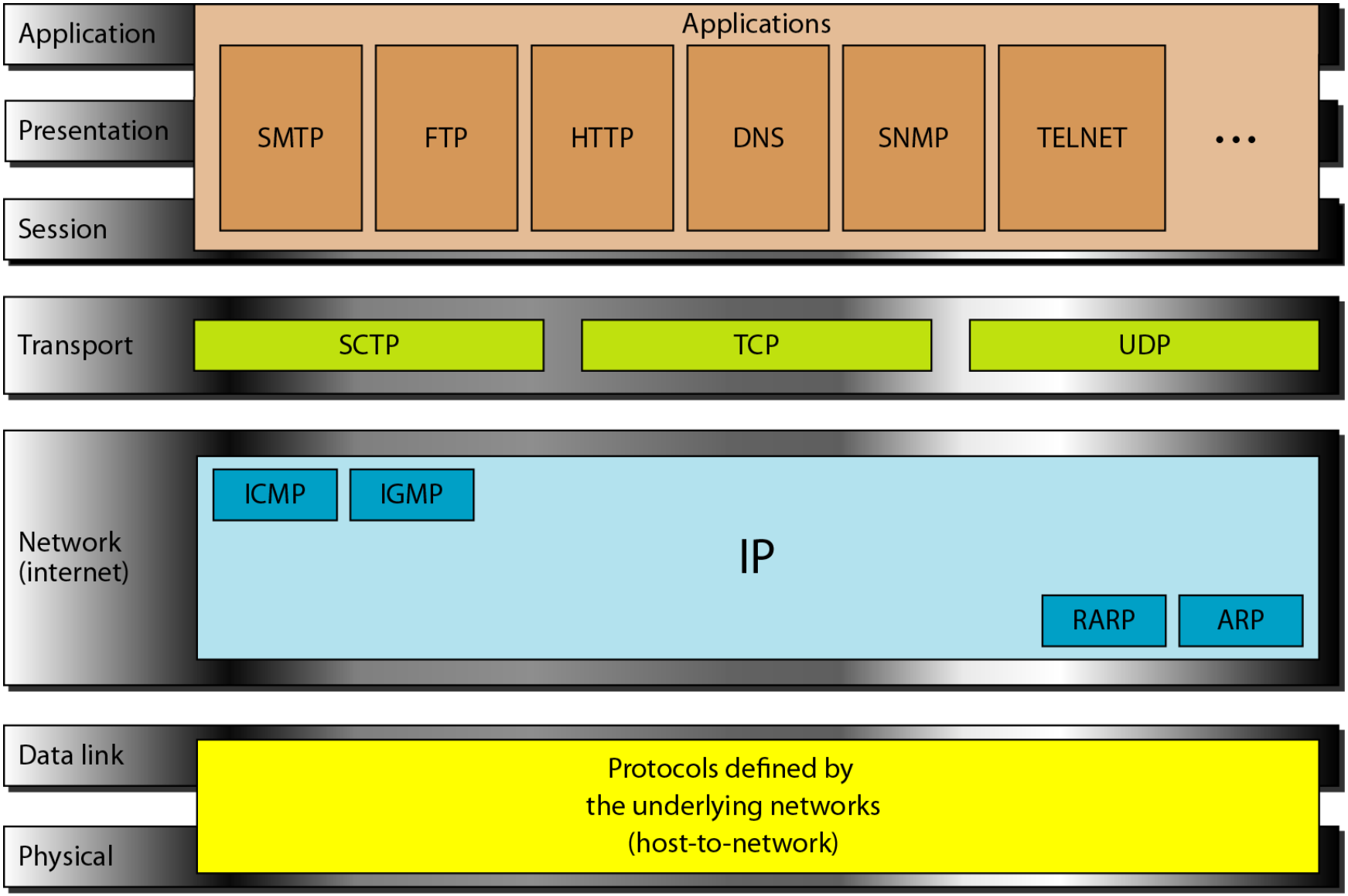
- TCPIIP protocol suite is made of five layers:
  - physical,
  - data link,
  - network,
  - transport, and
  - application

OSI & TCP/IP Models





TCP/IP and OSI model



- *TCP/IP* is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality.
- The term *hierarchical* means that each upper-level protocol is supported by one or more lower-level protocols.

- At the ***transport layer***, *TCP/IP* defines three protocols:
  - Transmission Control Protocol (TCP),
  - User Datagram Protocol (UDP), and
  - Stream Control Transmission Protocol (SCTP).

- At the ***network layer***, the main protocol defined by TCP/IP is the Internetworking Protocol (IP)

Physical and Data Link Layers

- At the physical and data link layers, *TCP/IP* does not define any specific protocol.
- It supports all the standard and proprietary protocols.

Network Layer:

- At the network layer, TCP/IP supports the Internetworking protocol.
- It in turns uses four supporting protocols:
  - ARP,
  - RARP,
  - ICMP,
  - IGMP.

Internetworking Protocol:

- It is a transmission medium used by TCP/IP protocols.
- It is an unreliable and connectionless protocol.
- IP transports data in packets called datagrams, each of which is transported separately.
- Datagram can travel along different routes and can arrive out of sequence or be duplicated.
- IP does not keep track of the routes and has no facility for reordering datagrams once they arrive at their destination

(No Error checking and Tracking)

- IP provides the following services
  - Fragmentation
  - Addressing

- **Fragmentation:**
  - Breaking the message into small pieces, if an intervening network cannot handle a large message
- **Addressing:**
  - Determining the route to deliver the data to the destination host. IP uses four supporting protocols
    - The Address Resolution Protocol (ARP)
    - The Reverse Address Resolution Protocol(RARP)
    - The Internet Control Message Protocol (ICMP)
    - The Internet Group Message Protocol (IGMP)

- **The Address Resolution Protocol (ARP)**
  - It is used to associate a logical address with a physical address.
- **The Reverse Address Resolution Protocol(RARP)**
  - It allows a host to discover its Internet address when it knows only its physical address.
- **The Internet Control Message Protocol (ICMP)**
  - It is a mechanism used by hosts and gateway to send notification of datagram problems back to the sender.
  - ICMP sends query and error reporting messages.
- **The Internet Group Message Protocol**
  - It is used to facilitate the simultaneous transmission

Transport Layer

- Traditionally the transport layer was represented in *TCP/IP* by two protocols:
  - TCP and
  - UDP.

- **IP is a host-to-host protocol**
  - meaning that it can deliver a packet from one physical device to another.

- **UDP and TCP are transport level protocols**
  - responsible for delivery of a message from a process (running program) to another process.

User Datagram Protocol

- The UDP is an unreliable connectionless protocol for applications

- It is a process-to-process protocol that adds only port addresses, checksum error control, and length information to the data from the upper layer.

Transmission Control Protocol

- TCP is a reliable stream transport protocol.
- The term stream means connection oriented. A connection must be established between both of ends of a transmission.
- At the sending end of each transmission, TCP divides a stream of data into smaller units called *segments*.
- Each segment includes a sequence number for reordering after receipt, together with an acknowledgment number for the segments received.

Stream Control Transmission Protocol

- The SCTP provides support for newer applications such as voice over the Internet.
- It is a transport layer protocol that combines the best features of UDP and TCP

Application Layer

- The *application layer* in TCP/IP is equivalent to the combined session, presentation, and application layers in the OSI model

Thank You